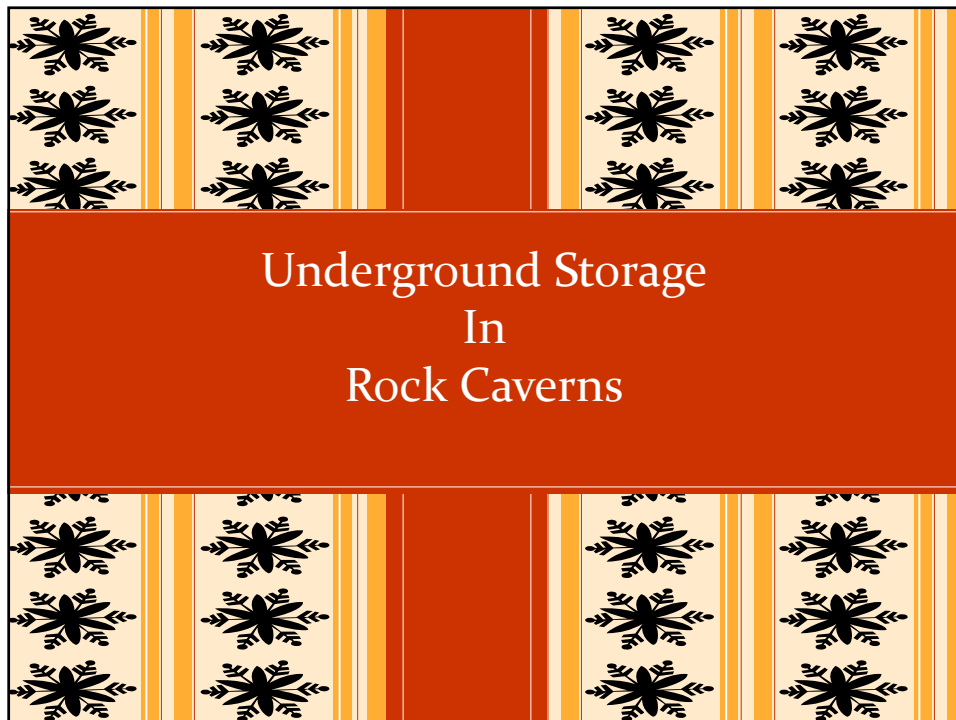
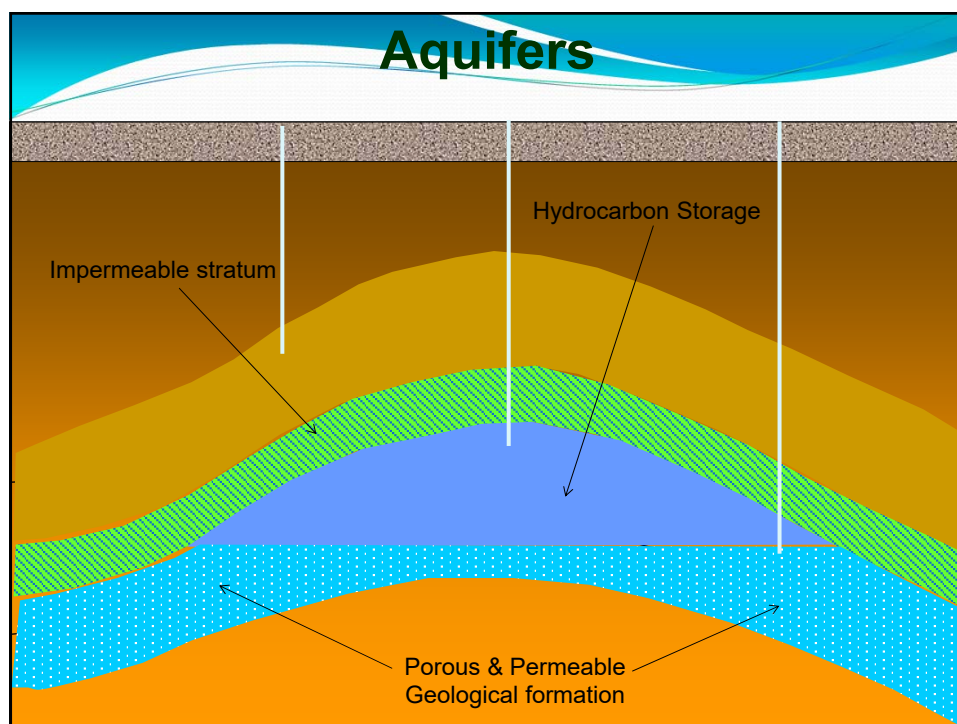
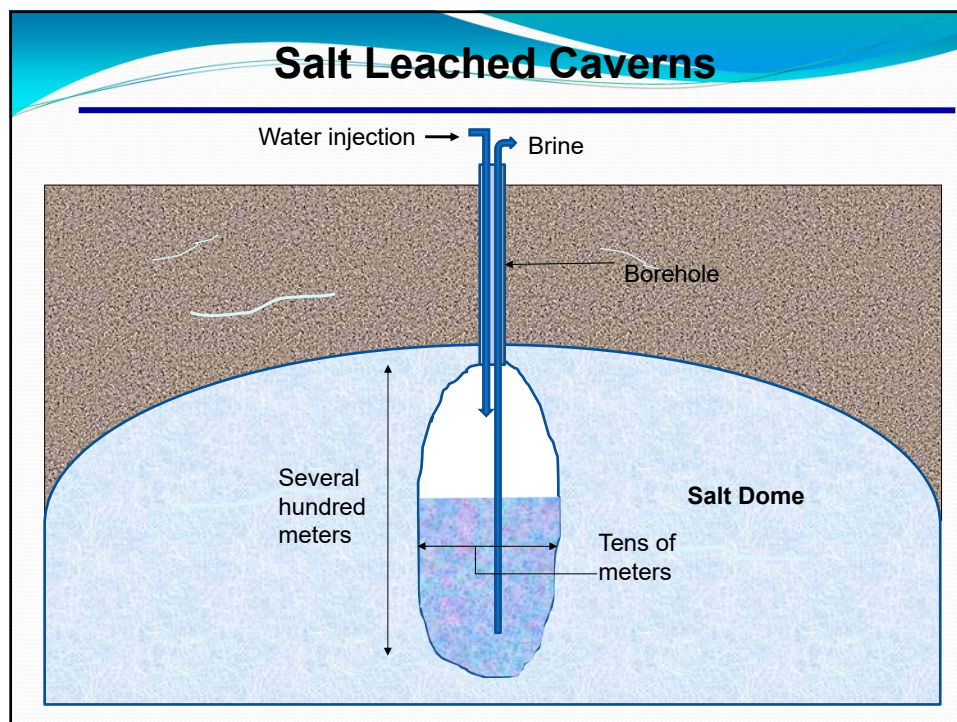
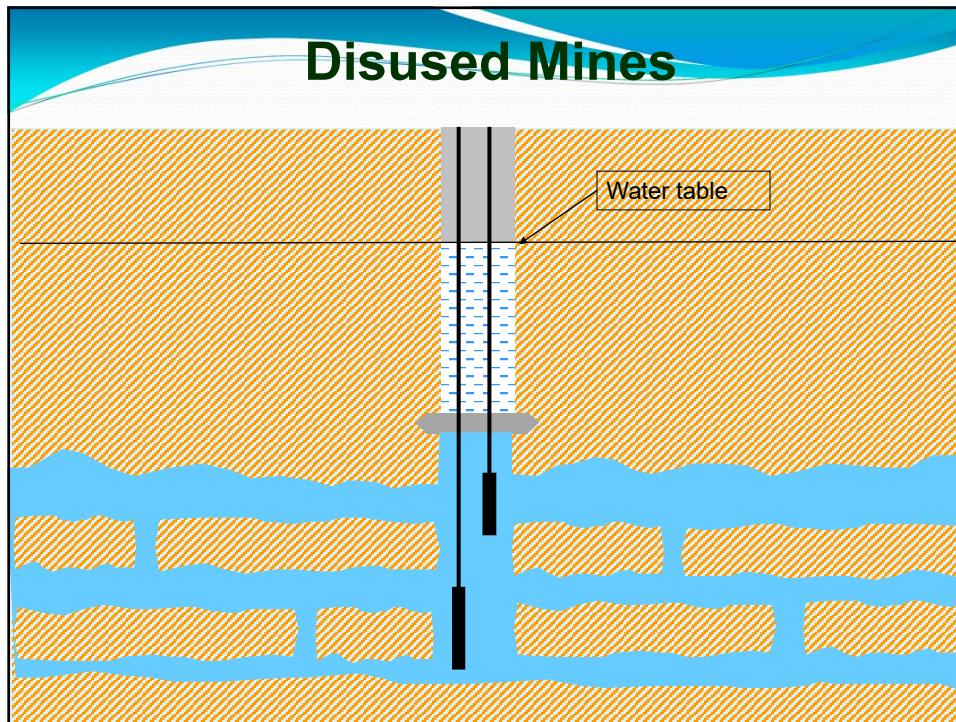




Types of Underground Storages

- Salt Leached Caverns.
- Aquifers or Depleted fields
- Disused mines.
- Underground Rock Caverns (Unlined & lined)





Types of Storages for Oil & Gas

- Conventional Storage
 - Constructed above ground
 - To meet short term storage & day to day operating needs of a Refinery
- Underground Storage
 - Strategic or commercial purpose
 - Un-lined Rock Caverns
 - Lined Rock Caverns
 - Depleted Reservoirs
 - Salt Caverns

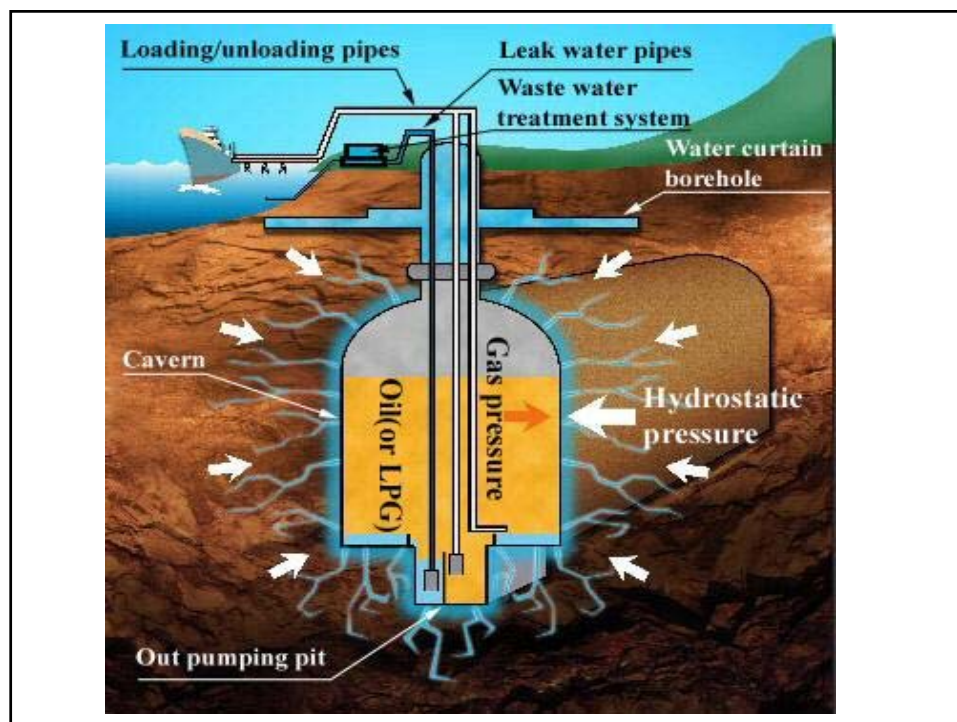
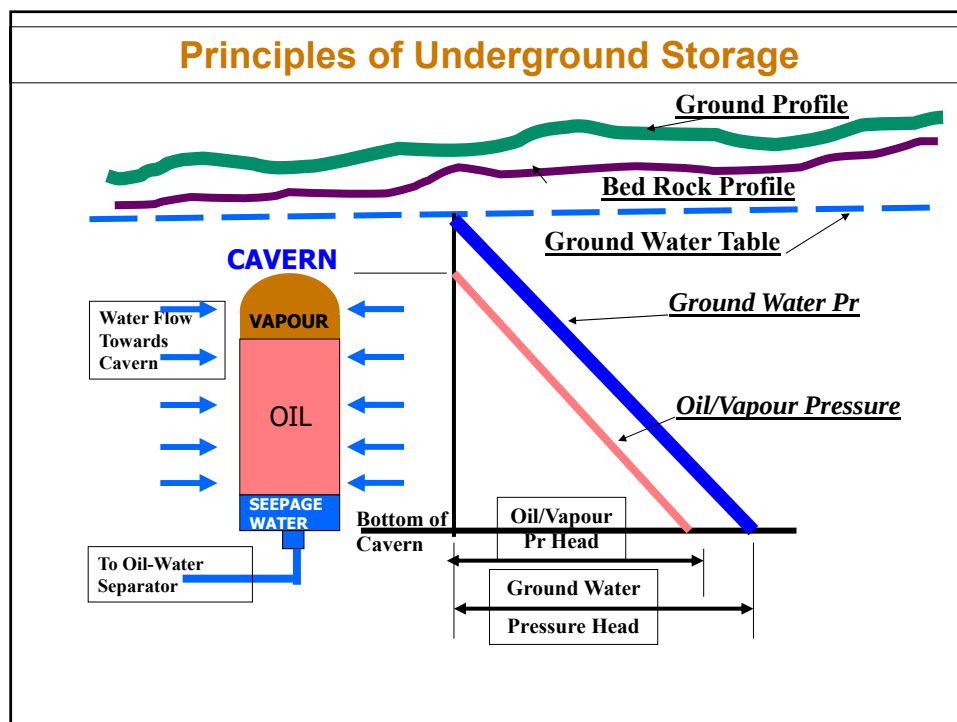
Advantages of Underground Storage

- To meet national demands in times of emergencies such as:
 - Supply disruption (wars, weather etc.)
 - Safety against enemy attack during wars
 - Price Fluctuations of Crude
 - Balancing of energy requirements
- Resistant to fire hazards
- Economical for larger capacities (commercial)

Underground unlined Rock Caverns

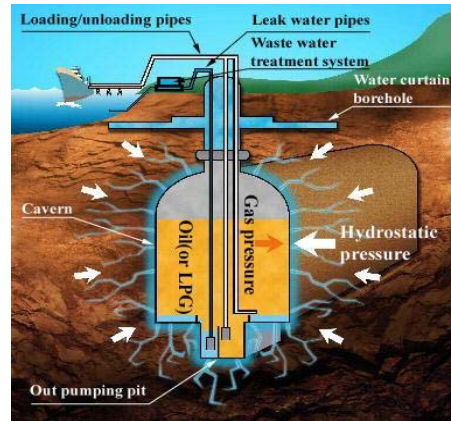
Basic Requirements

- Availability of Competent Rock at relatively shallow depths to ensure a stable cavern configuration
- High permanent Ground Water Table above top of cavern to ensure water flow always into cavern

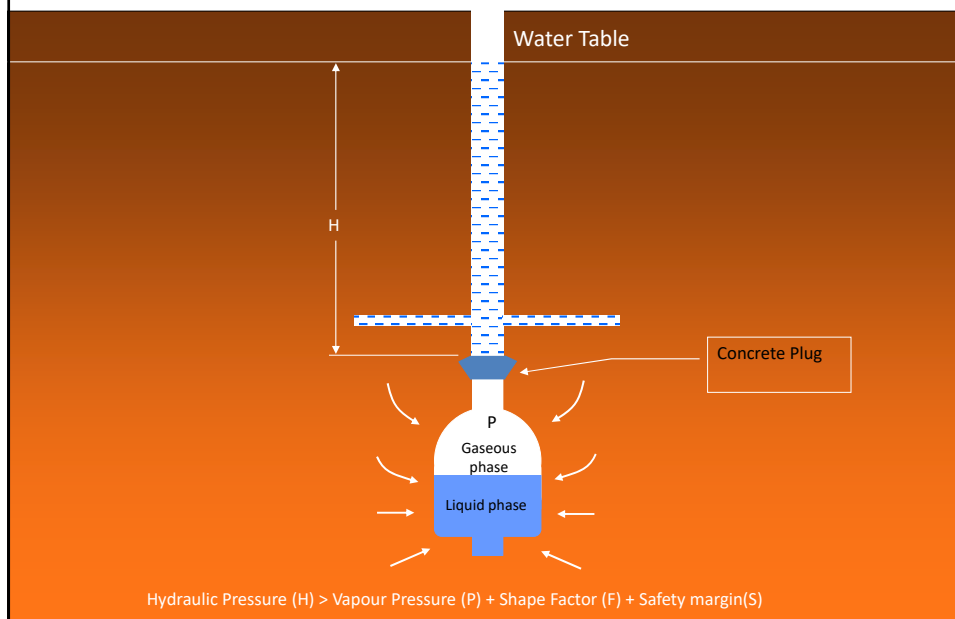


Unlined Rock Caverns (Schematic View)

- Principle of underground oil storage
 - Oil is lighter than water
 - Insoluble in water
 - Higher hydrostatic pressure prevents gas leakage and oil migration
- Gas Tightness Control Methods
 - Permeability controlling by grouting
 - Hydrodynamic containment by water curtain system
- Basic Requirements
 - Availability of Competent Rock at relatively shallow depths;
 - High permanent Ground Water Table;



Hydraulic Containment Principle



Hydraulic Containment Principle

- The rock caverns are planned at a depth such that there is sufficient hydrostatic pressure to counter the vapour pressure or liquid pressure of the stored product on the walls of the cavern
- These caverns are created where permanent ground water table is high. In order to further secure the water flow from the rock mass towards the cavern, a water curtain system is provided consisting of galleries located above the crown of the cavern.
- A saturated rock mass and ground water flowing into caverns, ensures proper sealing of the stored product from leakage. For ensuring stability of the caverns, a support system of shotcreting and rock bolting is commonly used.

Principles of Underground Storage

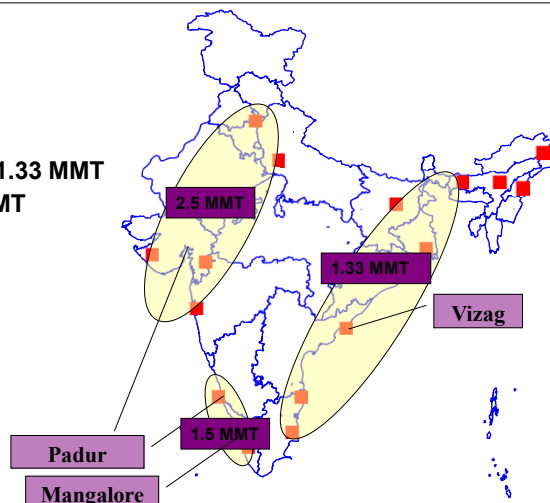
- The design of the storage is based on the **hydraulic containment principle**:
 - The hydrostatic pressure at cavern depth has to be sufficient to counterbalance the cavern pressure and confine the stored product in the underground storage cavern.
 - In order to check and secure the water flow from the rock mass towards the cavern a water curtain system consisting of galleries located 14 m above the crown of the cavern from which a number of horizontal boreholes are drilled in order the whole storage cavern area.

Principles of Underground Storage

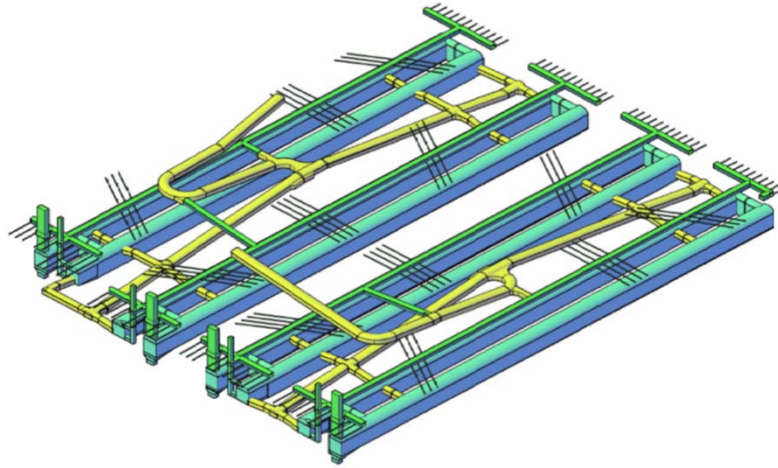
- This system is in use worldwide and many underground storages of crude oil (or LPG) based on this principle are operating successfully for more than 40 years.

Strategic Petroleum Reserves of INDIA

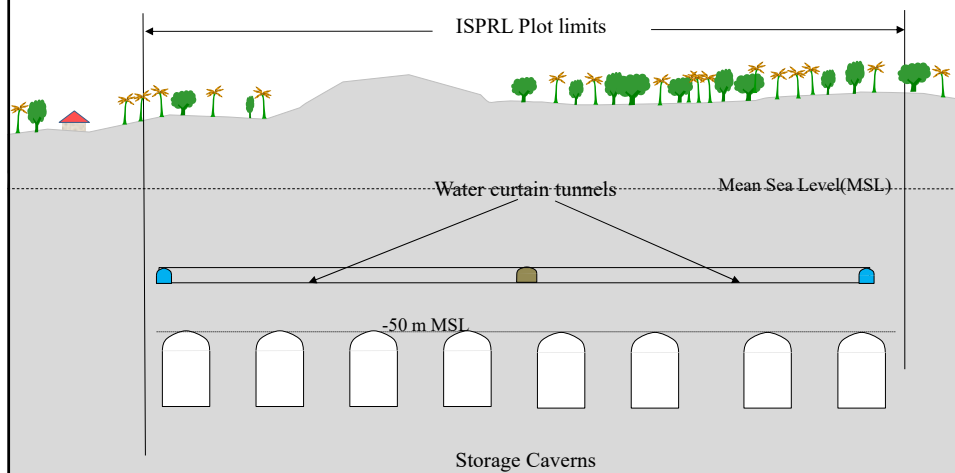
- **Locations:**
 - Vishakhapatnam : 1.33 MMT
 - Mangalore : 1.5 MMT
 - Padur : 2.5 MMT



Typical cavern layout with water curtain system



Caverns below the Ground



Components of Underground Storage
➤ STORAGE CAVERN (20x30m) ➤ ACCESS TUNNEL (12X8m) ➤ WATER CURTAIN TUNNEL (6.5x6.5m) ➤ BORE HOLES (100mm, 50-100m) ➤ SHAFTS (6x12, 4X4) ➤ CONCRETE PLUGS IN SHAFT AND TUNNEL ➤ CONCRETE FLOOR IN CAVERN ➤ CRUDE HOT OIL PIPE IN CAVERN FLOOR ➤ CASINGS AND PIPES IN SHAFTS & PUMP PIT

Projects Completed in India
▪ Vishakhapatnam ▪ Mangalore ▪ Padur, Karnataka ➤ Designed to store 5.00 M Tonnes of imported Crude (Equivalent to 15 Days Refining Throughput)

Components of Storage facility

- Storage cavern
- Access Tunnel
- Shaft
- Water curtain system:
 - Water curtain gallery
 - Water curtain boreholes

Storage Cavern

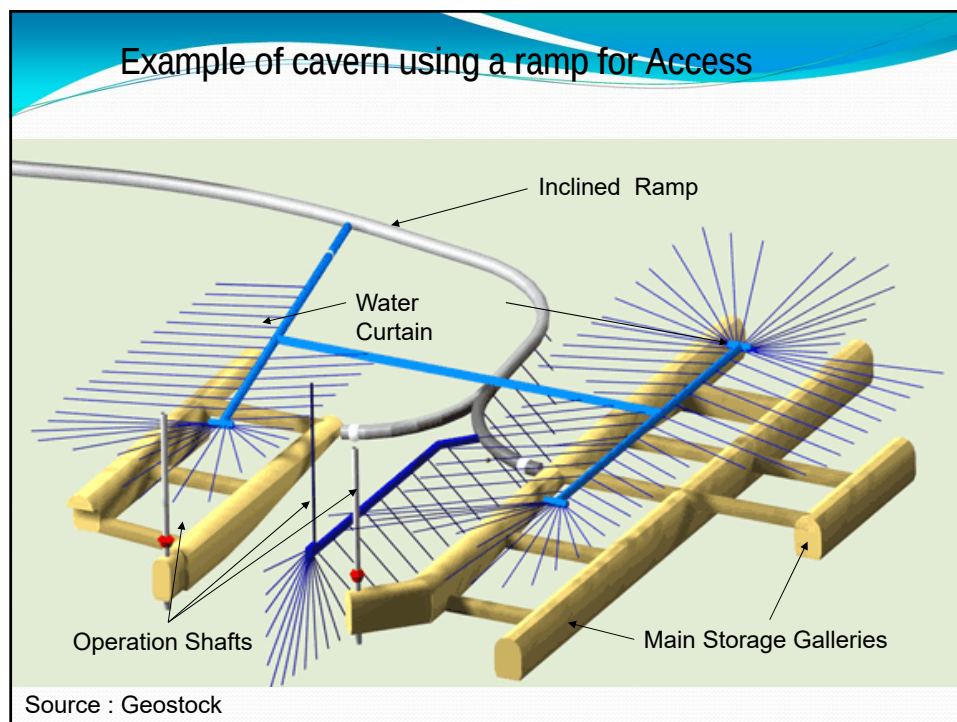
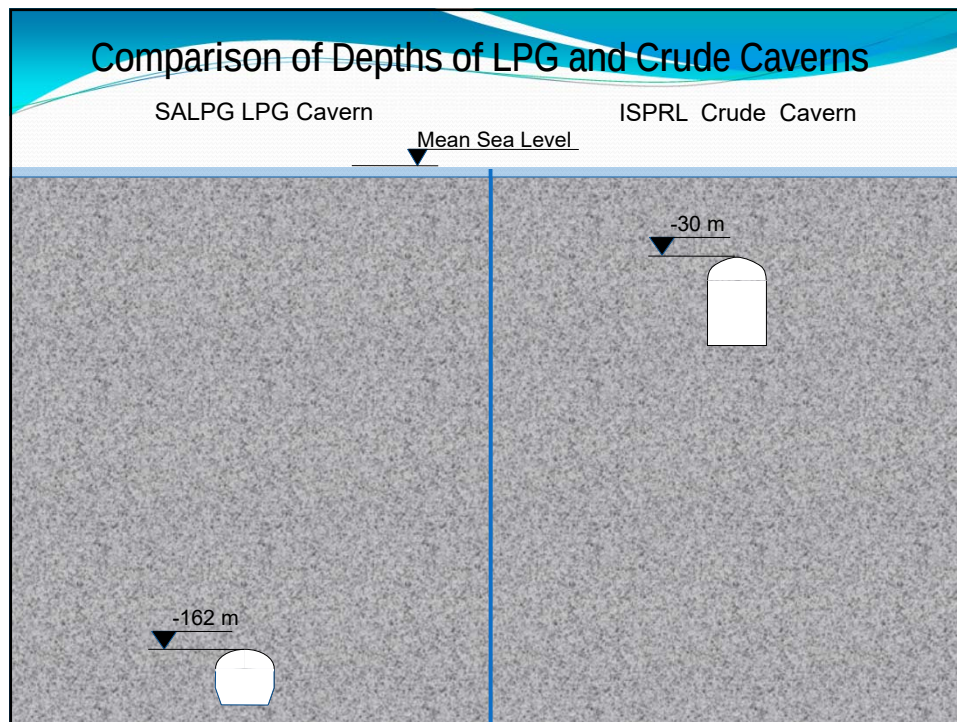
- Cavern shape at plan : U shaped
- Cavern Height : 30m
- Cavern width: 20m
- Cavern shape : D shaped
- Cavern Top: horizontal at same elevation
- Cavern Bottom: slope covered by concrete
- Cavern Inter distance: 30m for same product (Geotechnical)
- Cavern Inter distance: 60m for different product (Hydro geological)

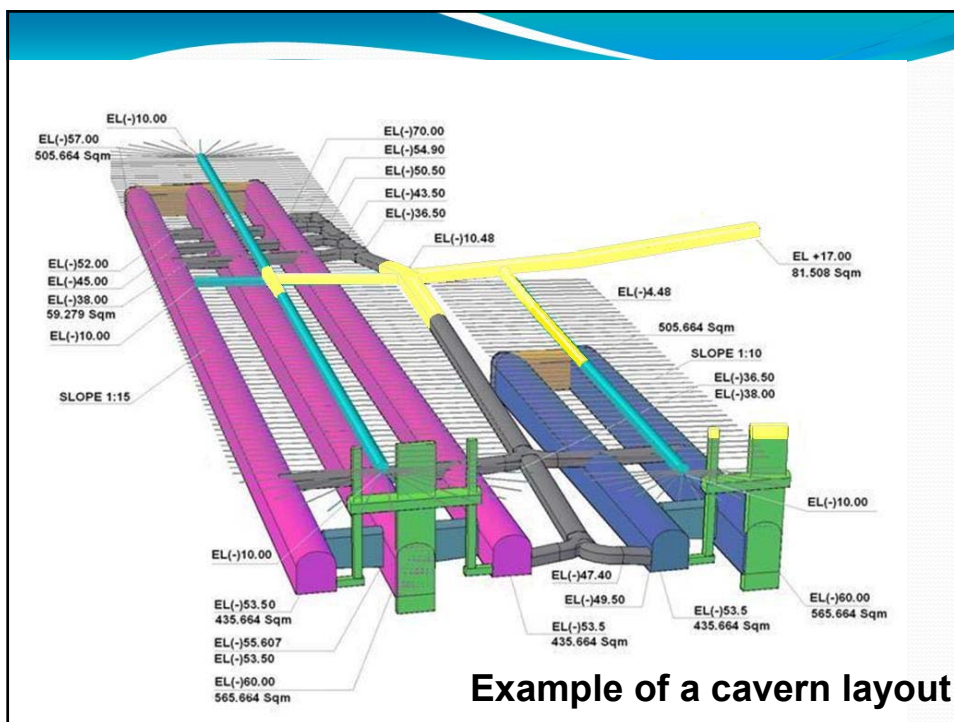
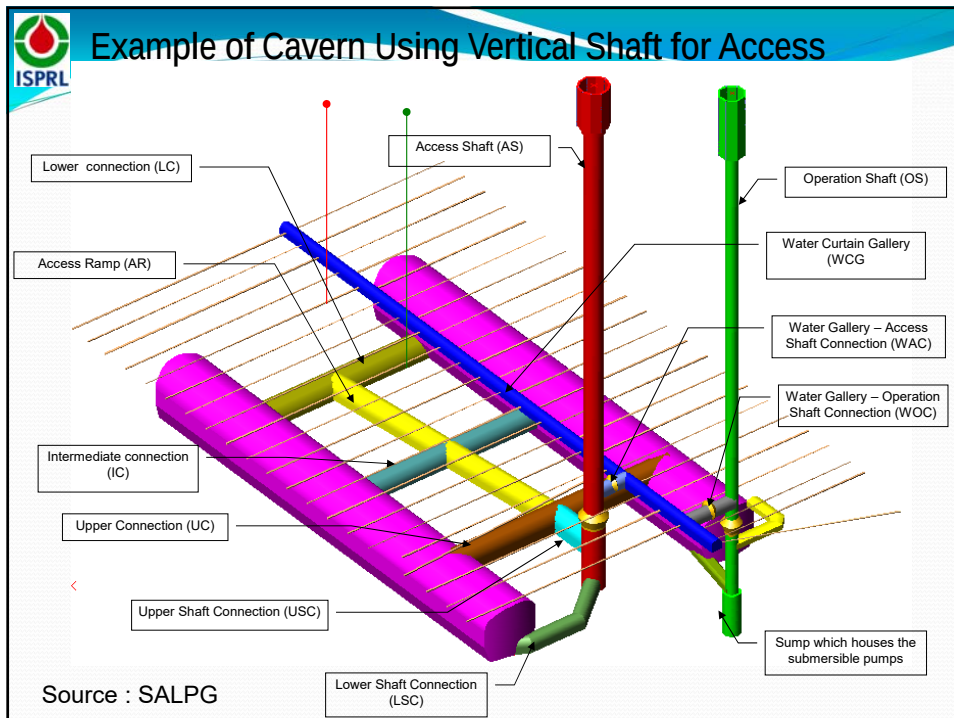
Access Tunnel & Shafts

- Access Tunnel
 - for construction purpose only
 - Size: As per construction plan (12 x 9m Typically)
 - Filled with water at a level close to natural GWL
- Shafts
 - For construction as well as Inlet & outlet for crude oil
 - Inlet shaft & outlet shaft per cavern unit
 - Pump pit at shaft bottom for submersible pumps to pump out water

Water Curtain

- **Water Curtain Gallery:** meant for drilling of water curtain boreholes
- Water curtain gallery size: Typically 6 x 6 m
- Water curtain boreholes:
 - Extension: 20m beyond storage cavern
 - Length: 50-100m
 - Spacing: 10-20m
 - Orientation: based on joint patterns in the rock mass
 - Diameter of boreholes: 95 -100 mm





Operation Principles and surface facility

- For avoiding explosive atmosphere in the vapour space in the caverns the product movement will require a gas compensation method based on inert gas production unit (withdrawal) and flaring (filling).
- In addition, in order to provide some flexibility in the vapour phase management the caverns are design for a pressure range of 0.1 to 1.5 bar g. (gauge) and the vapour phase of the two caverns will be interconnected.

Operation Principles and surface facility

- Considering the geothermal temperature at cavern depth (28 to 30 °C) and the properties of the Light and Heavy Arabian and Quaiboe crude oil, there is no requirement for product heating and it is anticipated that problems related with waxes or sludge accumulation, if any, will be of very limited importance.

Operation Principles and surface facility

However in order to cope for the storage of crude oil with different characteristics in the future the facility includes the following equipment and operation principles:

- a heating system in the sump of each operation shafts : it will consist in a coil with hot water heated at surface flowing inside a heater.
- the crude oil will be injected at an extremity of the cavern, opposite to the pumping pit
- a water bed of 30 cm is maintained at the bottom of the cavern; the heating line installed in the sump will be extended at the extremity of the cavern in order to inject and circulate hot water and chemical additives against bacteria formation. Heating system at surface will be installed only in the future, when required.
- A over excavation at invert level could also be considered.

Operation Principles and surface facility

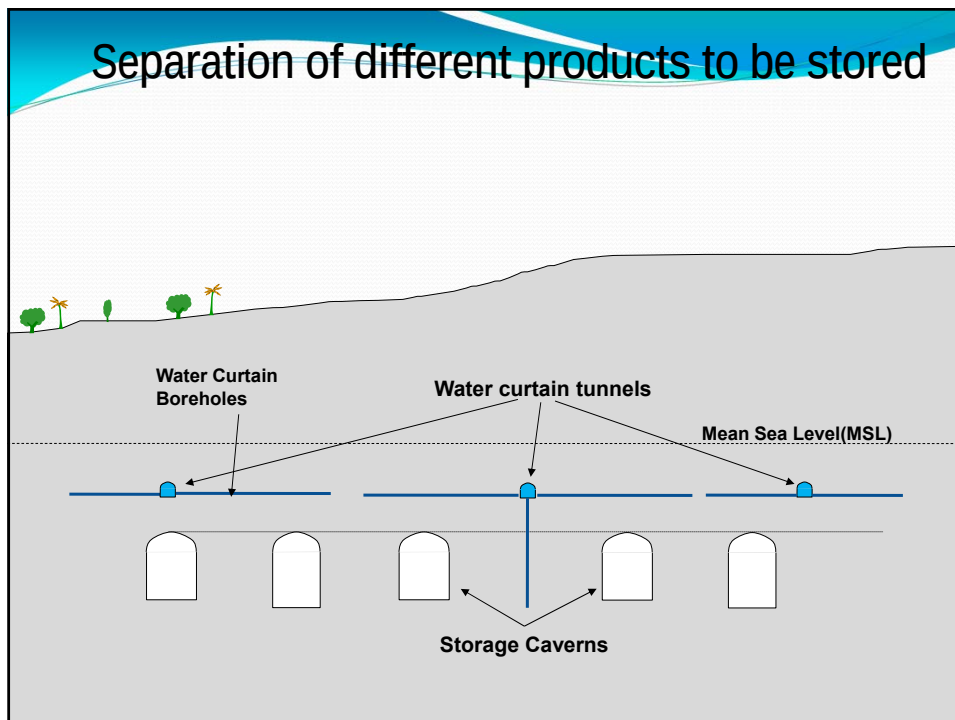
The above ground installations for operation of the two caverns and for reception and expedition by pipeline will be concentrated on the same plot for obvious economical, safety and environmental reasons. The main above ground facilities will consist of:

- Inlet/outlet metering
- Inert gas generator
- Flare
- API/CPI separator, seepage water treatment
- Shaft shelters
- Fire fighting pumps and water storage
- Voltage transformers
- Emergency generator
- Compressed air system
- Buildings (Administration/Control, electrical, maintenance/workshop).

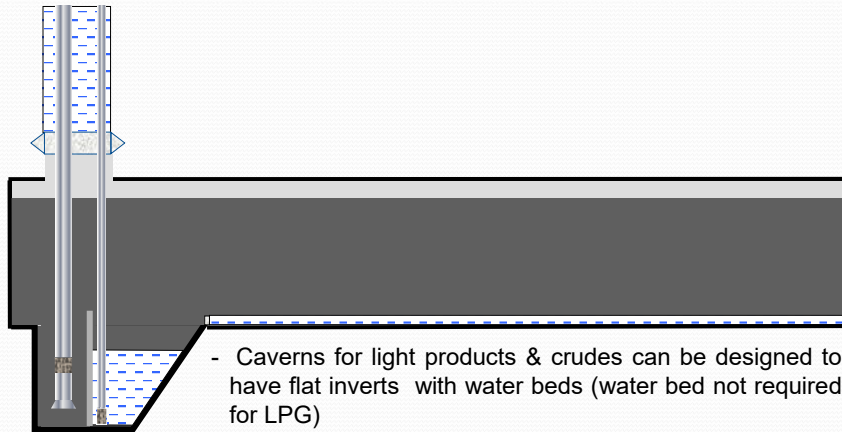
MAJOR UNDERGROUND WORKS-PADUR

- ❑ ACCESS TUNNEL OF 11(W) X 8(H) M AND LENGTH 2100 M
- ❑ WATER CURTAIN TUNNEL OF 6.5 X 6.5 M AND LENGTH 2800 M
- ❑ BORE HOLES OF LENGTH 50-100 M WITH DIAMETER 100 MM
- ❑ CONSISTS FOUR STORAGE UNITS OF 700 M LONG CAVERN OF 2.5 MMT
- ❑ THREE UNITS (TOTAL 1.875 MMT) FOR HIGH SULPHUR AND ONE UNIT OF 0.625 MMT FOR LOW SULPHUR
- ❑ CAVERN CROSS SECTION 20 (W) X 30 (H) M
- ❑ FOUR PUMP SHAFTS OF 6 X 12 M
- ❑ FOUR INLET SHAFTS OF 4 X 4 M

Separation of different products to be stored

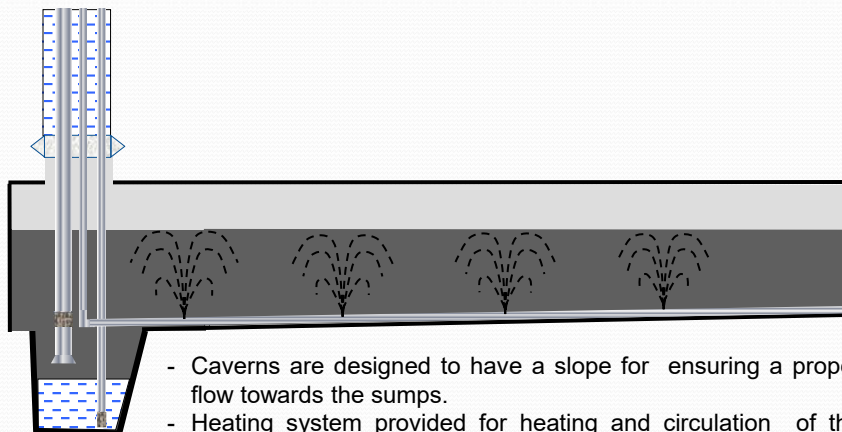


Design for light Products and Crudes (Flat Invert)



- Caverns for light products & crudes can be designed to have flat inverts with water beds (water bed not required for LPG)
- These caverns have no heating systems.
- Can be made for products which do not have major sludge problems

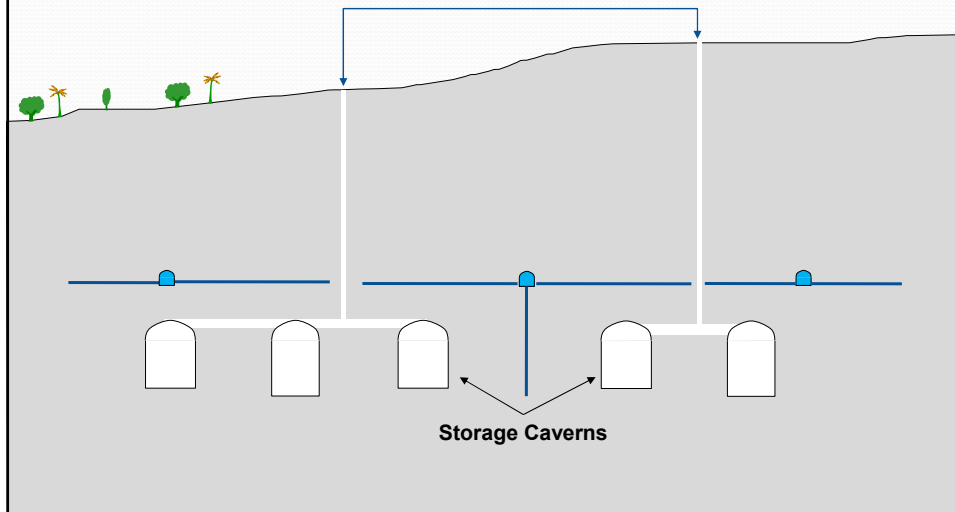
Design for Heavy Products and Crudes (Sloping Invert)



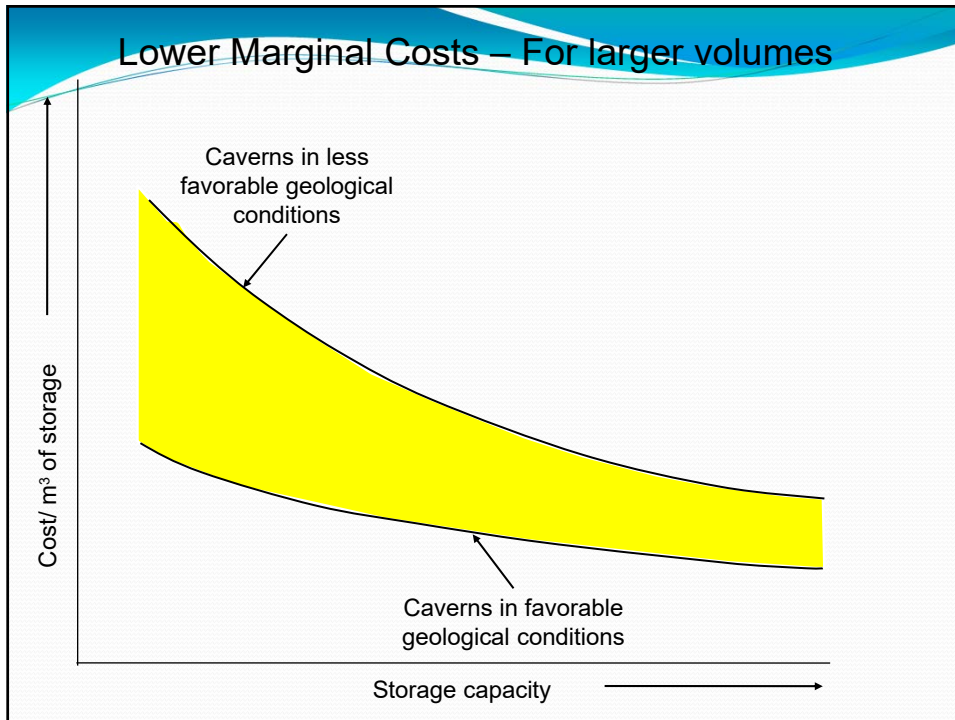
- Caverns are designed to have a slope for ensuring a proper flow towards the sumps.
- Heating system provided for heating and circulation of the product stored.
- The design permits storage of heavy oils and heavy crudes.

Inertisation of caverns (products other than LPG)

Connecting the vapour spaces can reduce the requirement of inert gas.



Lower Marginal Costs – For larger volumes



Cavern Design (PADUR)

- Facility designed for two crude qualities :
 - ◆ High Sulphur
 - ◆ Low Sulphur
- Consists of two separate storage units in 75 : 25 ratio
- Cavern layout : tunnels with “D” shaped cross section. Product intake located at one end of the “leg” ; Pump installation located at end of other “leg”.
- Layout / shape of caverns will undergo changes based on results of supplementary investigations.

Cavern Design contd...

- Cavern roof is horizontal along full length
- Cavern volume includes design volume of crude *plus* 3% extra for “vapour phase”.
- Extra volume provided equivalent to three days seepage water .
- Cavern floor sloped longitudinally 1 : 250 towards pump pit. Lateral slope 1 : 10 from middle towards sides.

Cavern Design contd...

- Cavern roof 50 - 70 M below ground level
- Water curtain located 15 m above cavern roof

The design will undergo change after basic engineering is complete

Cavern Design contd...

- The layout, cross-section and elevation of the caverns are designed considering the product to be stored and operational requirements as well as the geotechnical conditions and geological setting at site.
- The overall layout and the cross-section of the cavern is selected so as to achieve a favourable stress condition vis-à-vis the in-situ stress regime of the site which also take into account any major geological structures.
- The caverns are oriented in a direction so as to come across minimum geological problems such as joints, shear zones and principal planes of horizontal stresses in the rock while excavating the caverns.
- The caverns, in general have a D shaped cross section

Cavern Design contd...

- Estimation of the rock mass properties and rock support are initially performed using the empirical rock mass classification schemes proposed by Bieniawski (RMR), Barton (Q-system) and / or Hoek-Brown (GSI). The rock caverns are then numerically analysed for stresses and wedge stability.
- Depending on the site conditions, while the size and shape of the caverns could vary, the crown level of the caverns is designed adequately below the ground water level so as to ensure the hydro static pressure requirement for containment of the product. Hydrodynamic containment is further reinforced by provision of a water curtain system above the cavern. This is then further confirmed through numerical modelling.

Cavern Design contd...

- For the storage purpose, in general 'U' shaped caverns are planned, having leg lengths proportional to volume of the type of crude to be stored and number of caverns depending on the suitability of the site.
- While the cavern geometry is dependent on the geological setting of the selected area, in general the typical dimension of a cavern cross section is about 20m to 35m in height and 15m to 20m in width. However, the length of the cavern is designed to accommodate the requisite volume of storage and the extent of suitable rock quality at required depth is available.
- In general approach, the caverns are laid out parallelly with minimum separation of 30m.

Cavern Design contd...

- Water infiltration gallery and water curtains, above the crown level of the cavern are provided to ensure a constant water pressure towards the caverns. Boreholes are drilled from the water curtain tunnel to intersect all the joints of rock mass. The water curtain gallery and boreholes are required to be pressurised at least 50m ahead of the main cavern to ensure saturation during the construction phase.
- The submersible pumps in the caverns are designed to deliver the crude oil either to the nearest refineries or back loading in the tankers for despatch to other refineries, through new pipelines to be connected to the existing pipelines.

FEATURES OF THE STORAGE FACILITIES (PADUR)

- Storage Capacity 2.50 MT
- Eight Parallel Caverns (700m)
- Unit 1 : Six Caverns for High Sulphur
- Unit 2 : Two Caverns for Low Sulphur
- Cavern size up to 20m span and 30m height
- Water curtain holes of 50 to 100 m length at 10m intervals

Constructional Aspects

Tunneling:

- Access Tunnel sizing
- Drill & Blast cycle
- Ventilation
- Scaling
- Mucking and disposal
- Rock support
 - Rock Bolting
 - Shotcreting
- Grouting

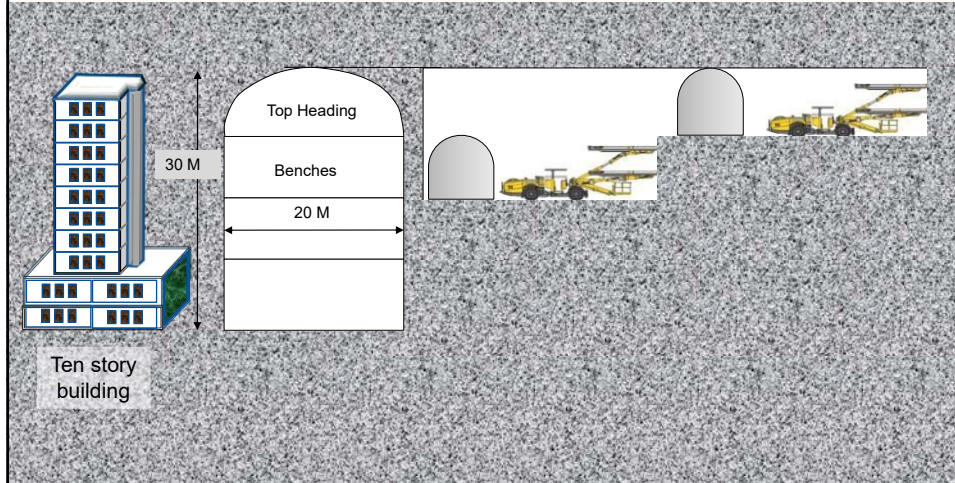
Excavation Sequence

Excavation of storage considered in 4 stages:

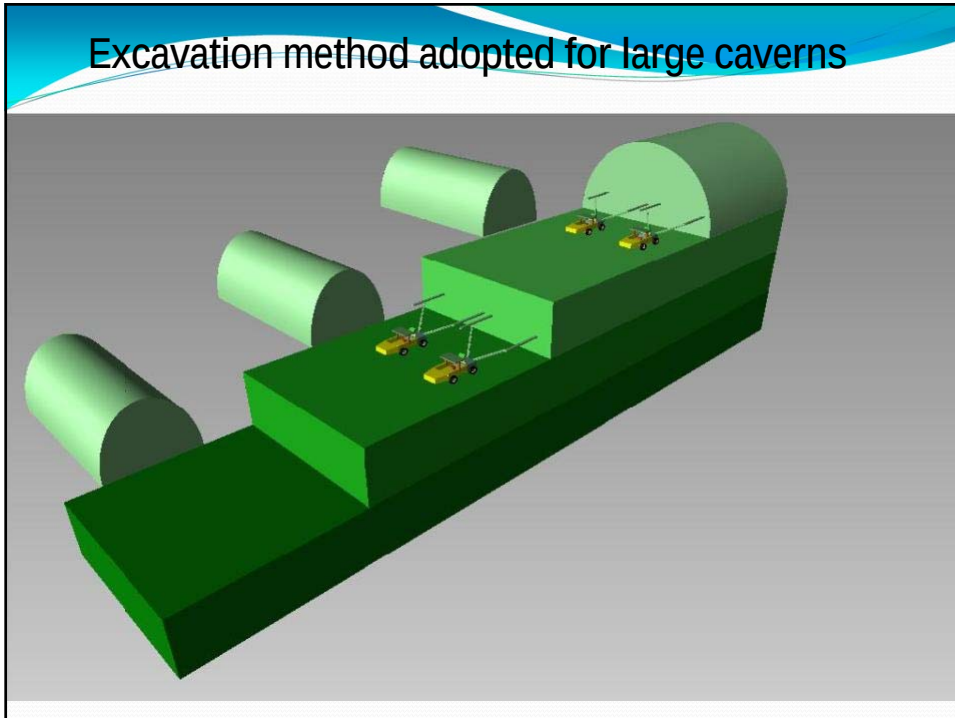
- ◆ Top heading
- ◆ 1st Bench
- ◆ 2nd Bench
- ◆ 3rd bench

Large size- Need for benching

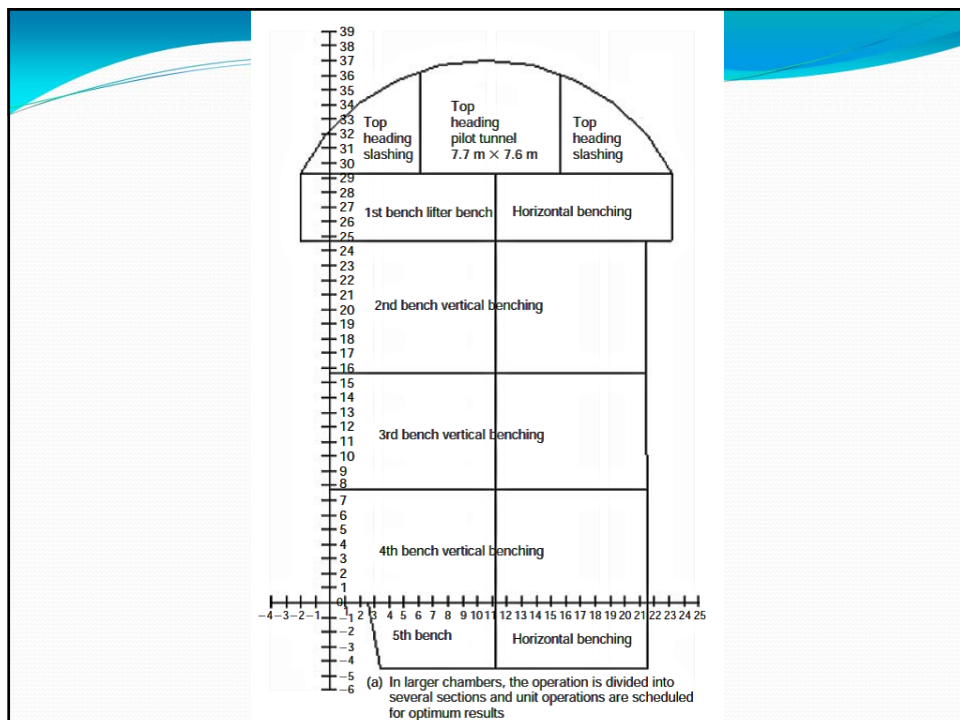
For large caverns there is a need to excavate the caverns in steps. The top heading is completed first followed by benching

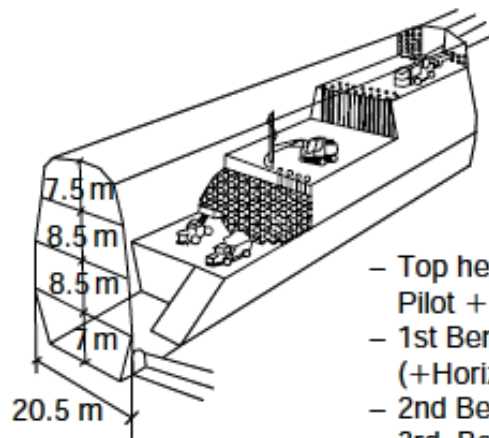


Excavation method adopted for large caverns



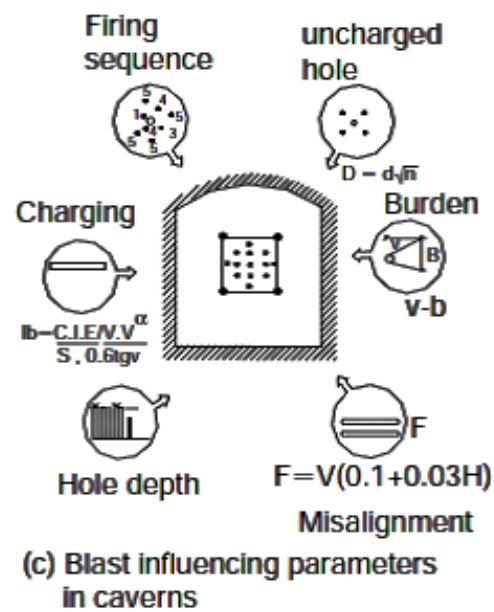
Need for Access to Various Levels in Large Caverns





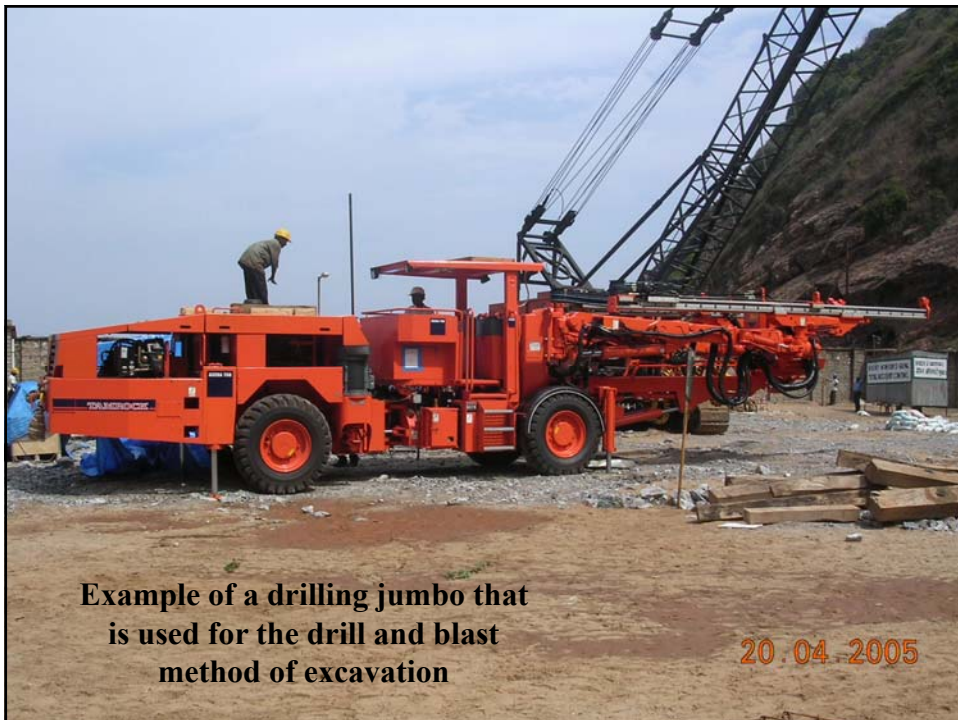
- Top heading 7.5 m = Pilot + Reaming
- 1st Bench vertical 8.5 m (+Horizontal reaming)
- 2nd Bench vertical 8.5 m
- 3rd Bench horizontal/vertical

(b) Pictorial view showing unit operations while creating large underground chambers/excavations



Excavation Using Drill & Blast (For Hard Rock)

Two Boom Jumbo (Used for drilling in hard rock)



Example of a drilling jumbo that
is used for the drill and blast
method of excavation

Drilling & Blasting Method of Excavation

Jumbo used for
drilling holes in rock



Holes are charged with
explosives after drilling
and controlled blasting
carried out

Drilling & Blasting Method of Excavation

Pull

Vibrations During Blasting

Construction of underground caverns involves extensive blasting operations for tunneling works.

Controlled and smooth blasting techniques would be employed to minimize any risk of damage to any neighbouring structures.

Concrete Works

- Tunnel Portal
- Concrete Plugs in Cavern Shafts & Access Tunnels
- Cavern Floor
- Around Circulation Pipes on Cavern Floor
- Sump Pit; Base of equipment and pipe supports
- Buildings

Water Curtain System

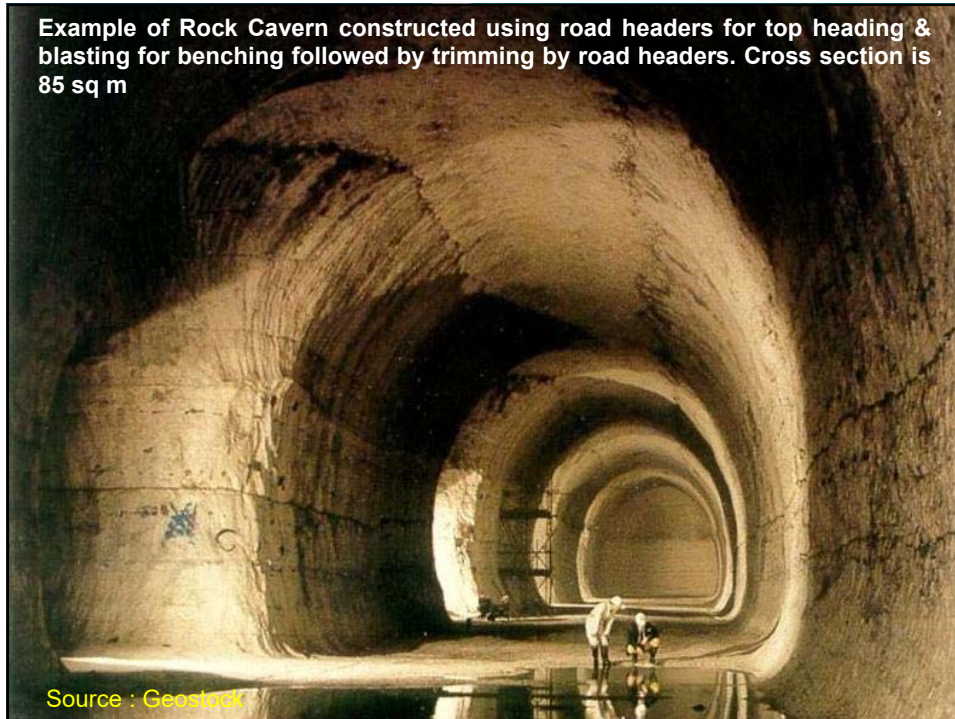
- **Function:** To maintain water pressure in fissures above storage cavern.
- Water Curtain Tunnels: (*design may undergo change*)
 - ◆ Driven from access tunnel
 - ◆ Located across centre line of the caverns
 - ◆ Floor elevation approx. 15 m above cavern roof level.
 - ◆ Size 6.5m X 6.5m, four nos. required.
- Boreholes for water curtain;
 - ◆ Drilled from water curtain tunnel @ 10m spacing
 - ◆ Drilled at slightly downwards inclination.
 - ◆ To extend 20 m outside storage cavern area.

Excavation using Road Headers (For Soft Rock)

Road Headers (Used in case of soft to medium hard rock Limestone, Chalk etc)

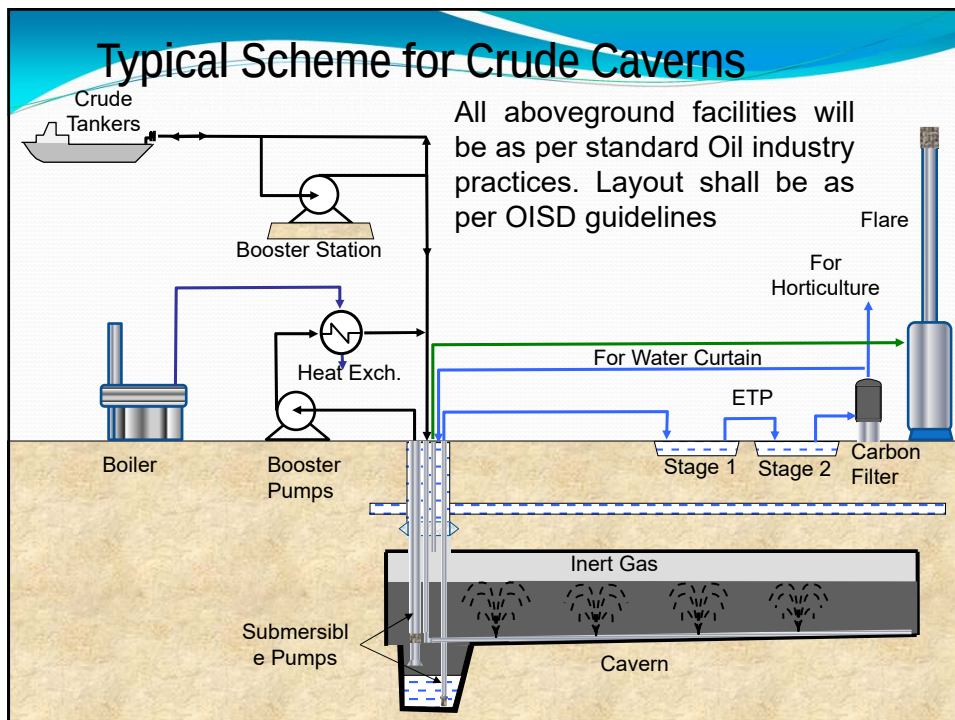


Example of Rock Cavern constructed using road headers for top heading & blasting for benching followed by trimming by road headers. Cross section is 85 sq m



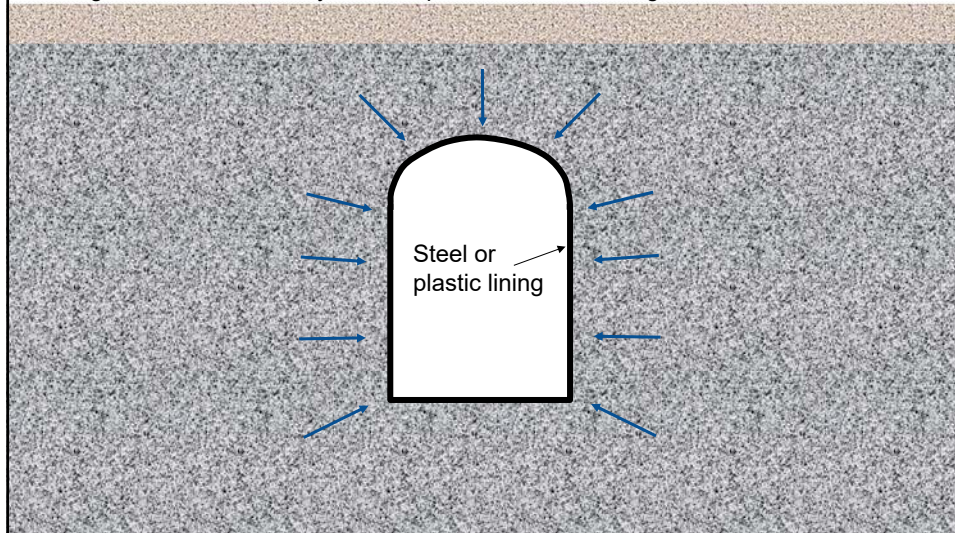
Source : Geostock

Typical Scheme for Crude Caverns



Lined Rock Caverns

These types of storage facilities can be used for small or medium sized caverns. The lining can be steel or plastics. In the depths are large the linings have to be designed to counter the hydrostatic pressure or the can get



Design and Construction of Caverns

- A rock cavern is a man- engineered cave, for a specific application.
- There are over 20,000 caverns built around the world, for a variety of applications, ranging from storage of oil and gas to sport and concert halls.

Design and Construction of Caverns

Suitable Geology

- Rock caverns are generally unsupported openings. They are generally constructed in competent rock masses so the rock masses can be self-supported.
- Most caverns are constructed in granitic and crystalline rocks. Limestone and strong clastic sedimentary rocks are also possible hosts.

Design and Construction of Caverns

Basis of Design

- a) The rock is used as a structural material.
- b) Geotechnical design is based primarily on precedents, i.e. empirical methods.
- c) The design is related to construction procedures.
- d) The design is optimised on the basis of rock mechanics, construction methods and usage, etc.
- e) Numerical methods can be used to predict problem areas and to extrapolate experience.
- f) Monitoring used to verify the design.

Design and Construction of Caverns

Design Sequence (i)

- a) Identification of the geometrical and physical requirements for the cavern.
- b) Identification of areas with geology suitable for cavern construction.
- c) Evaluation of the topography in relation to the geometrical requirements.
- d) Location of suitable access to the underground facility.
- e) Evaluation of geological and hydrogeological data.

Design and Construction of Caverns

Design Sequence (ii)

- a) Determination of optimal location, orientation, layout and geometry for the cavern or cavern system based on the above factors.
- b) Optimisation of the design with respect to cavern use and construction methods, which may include modification of the cavern use.
- c) Evaluation of rock support measures.

Design and Construction of Caverns

Design Consideration on Location and Orientation

- a) Adequate rock cover.
- b) Avoid weakness zones.
- c) Cross weakness zones in the shortest possible distance.
- d) Avoid adverse orientation relative to major joint sets.
- e) Make favourable use of groundwater pressures.
- f) Avoid rock with abnormally low stresses, or with very high stresses.

Design and Construction of Caverns

Minimum Rock Cover (i)

The rock cover should be sufficient so that the roof and walls will be self-supporting. The minimum rock cover is determined from many factors:

- a) the quality of the geological information and the rock properties,
- b) thickness of superficial deposits and depth of weathering,
- c) the cavern span and,
- d) cost implications.

Design and Construction of Caverns

Minimum Rock Cover (ii)

- As a general rule, the minimum cover of strong rock should be not less than half the cavern span.
- In general, reduced cover increases the amount and cost of ground investigation and rock support work and this cost must be offset by advantages in adopting reduced cover. Reduced rock cover is normally limited to small areas, such as the section of cavern closest to the portal.

Design and Construction of Caverns

Weakness Zones

Weakness zones can be formed by weak rocks, faults and deeply weathering, with thickness from a few centimetres to several hundred metres. In dealing with weakness zones,

- a) weakness zones must be identified,
- b) if possible, avoid weakness zone,
- c) minimise excavation in weakness zone,
- d) consider the orientation of the weakness zones.

Design and Construction of Caverns

Joints

- The orientation of joints with respect to the axis of the excavation influence the stability of a cavern.
- The orientation of joints influence the amount of overbreak.
- Optimization of excavation direction with respect to joint orientation can be achieved. E.g., the longitudinal axis of the cavern is ideally oriented normal to the line of intersection of the two dominant joint sets.

Design and Construction of Caverns

Groundwater

The location of the groundwater surface and predictions of changes created by the underground openings can be important considerations in determining the elevation of a cavern scheme.

- a) Groundwater inflow can be problem for excavation.
- b) Most cavern applications need dry environment.
- c) Water curtains are used to confine the oil and gas in caverns.

Design and Construction of Caverns

In Situ Stresses

In situ stresses influence the stability of excavations.

- a) In general, increased stresses give increased stability.
- b) Excessive high in situ stresses influence can cause strength failure of cavern.
- c) Stresses in hard rocks are normally anisotropic, can influence cavern stability.
- d) For high stresses, cavern section shape can be optimised.

Design and Construction of Caverns

Cavern Layout and Shape (i)

The design of cavern geometry and layout of a system of caverns is normally based on:

- a) Requirements given by the cavern usage.
- b) Costing of excavation and support operations.
- c) Geometry of the opening, i.e. the total height and arch shape, influences the cost of excavation and support.

Design and Construction of Caverns

Cavern Layout and Shape (ii)

- The main parameters defining cavern layout and geometry are the cavern size and shape and the spacing between caverns. They are primarily based on empirical guidelines from previous experiences.
- Large span caverns, caverns in difficult ground conditions and multi-cavern schemes are commonly subjected to stability and stress distribution analyses using various methods.

Design and Construction of Caverns

Design of Cavern Shape (General)

Rock mass is discontinuous of low tensile strength. The design of shape is to evenly distribute the compressive stresses in the surrounding rock mass:

- a) Use an arched roof;
- b) Avoid intruding corners;
- c) Optimise cross-section sizes to the lowest combined excavation and support costs;
- d) Optimise cross-section shape to the best stress distribution.

Design and Construction of Caverns

Design of Cavern Shape (Roof)

Roof arch is generally designed to have height:span of 1:5, and,

- a) The roof shape is not commonly altered to suit particular geological structures;
- b) Height may be reduced if the dominant joints have a shallow dip;
- c) Usually height will not be increased as economically not justified;
- d) Increasing the roof arch height only if the space under the arch for ducts and services is needed.

Design and Construction of Caverns

Design of Cavern Shape (Wall)

Cavern walls are normally vertical. Wall stability is a function of wall height, the in situ stresses and the orientation and properties of the principal joint sets.

- a) The flat wall surface has no arching action and high walls tend to be unstable;
- b) Major joints and seams can dominate wall stability and can affect the chosen wall height;
- c) The cost and scale of stabilising measures can increase substantially with wall height;

Design and Construction of Caverns

Design of Cavern Shape (Wall)

- d) Joints with shallow dip favour wall stability as the dominating vertical stresses in the walls increase joint friction;
- e) Steeply dipping joints with strikes parallel to the wall reduce stability as the horizontal stresses on the joints are small.

Design and Construction of Caverns

Design of Cavern Shape (Stress)

Anisotropic and high stresses may have to be taken into account in cavern design.

- a) For exceptionally high stresses, the shape of the cross- section should be optimised;
- b) Optimisation of shape can be analysed based on stress condition;
- c) There are cases that cavern cross-section reshaped due to anisotropic high stresses.

Design and Construction of Caverns

Design of Cavern Horizontal Spacing (Pillar Width)

Pillar width depends primarily on the rock quality, the discontinuity orientation, the cavern spans and heights.

- a) Pillar widths are normally equal to 0.5–1.0 full cavern span or height, whichever is the greater;
- b) Pillar widths are normally determined on the basis of judgement and simple analysis, e.g., possible sliding on unfavourable joints;
- c) Narrow pillars may be necessary because of site availability and other factors.

Design and Construction of Caverns

Design of Cavern Vertical Spacing (i)

Vertical separation in generally should not be less than span or height. It depends on the rock quality, the orientation of the discontinuities, the cavern dimensions, and in situ stresses.

- a) It generally requires detail analysis and modelling;
- b) It should consider overbreak and loosening of rock in both upper and lower caverns;
- c) It should consider the risk of outfall of rock may cause stability of upper cavern;

Design and Construction of Caverns

Design of Cavern Vertical Spacing (ii)

- a) It should consider the cost for blast and support;
- b) The stability of the separating rock may be improved by pre-grouting and bolting from either the upper or lower cavern;
- c) Excavation of the upper caverns before the lower caverns is recommended. This avoids the risk of damage to the roof support installed in the lower cavern by vibrations from the heavy charges used in the bottom of the upper caverns.

Design and Construction of Caverns

Basis of Cavern Support Design

- a) The rock is used as a structural material, i.e., primarily reinforcement
- b) Support design is based rock mass quality and precedents, i.e. empirical methods
- c) Numerical methods can be used to predict problem areas and to extrapolate experience
- d) Monitoring used to verify the design

Design and Construction of Caverns

Cavern Support Design Approach

- Preliminary design of rock support may be based on rock classifications (Q or RMR), to provide the most suitable types of support for the various rock classes that have been identified.
- Temporally reinforcement (often bolts and shotcrete in hard rock tunnelling) applied immediately after excavation can also serve as permanent reinforcement.
- Further permanent reinforcement is applied (bolts and shotcrete) later.

Design and Construction of Caverns

Cavern Support Design

Roof: Use Q-support design chart directly.

Wall:

For $Q > 10$,	$Q_{\text{wall}} = 5 Q$
For $0.1 < Q < 10$,	$Q_{\text{wall}} = 2.5 Q$
For $Q < 0.1$.	$Q_{\text{wall}} = Q$

Design and Construction of Caverns

Cavern Support Design

Temporary support: use the following adjustment,
Increase ESR to 1.5 ESR or
Increase Q to 5 Q (applicable to roof and wall).

Maximum unsupported span = $2 \text{ ESR } Q^{0.4}$

Example: Q = 10, ESR = 1, maximum unsupported span = 5 m

Design and Construction of Caverns

Construction Method

Rock caverns are generally excavated by drill-and-blast method, and supported by bolts and shotcrete.

Cavern excavation is usually done by:

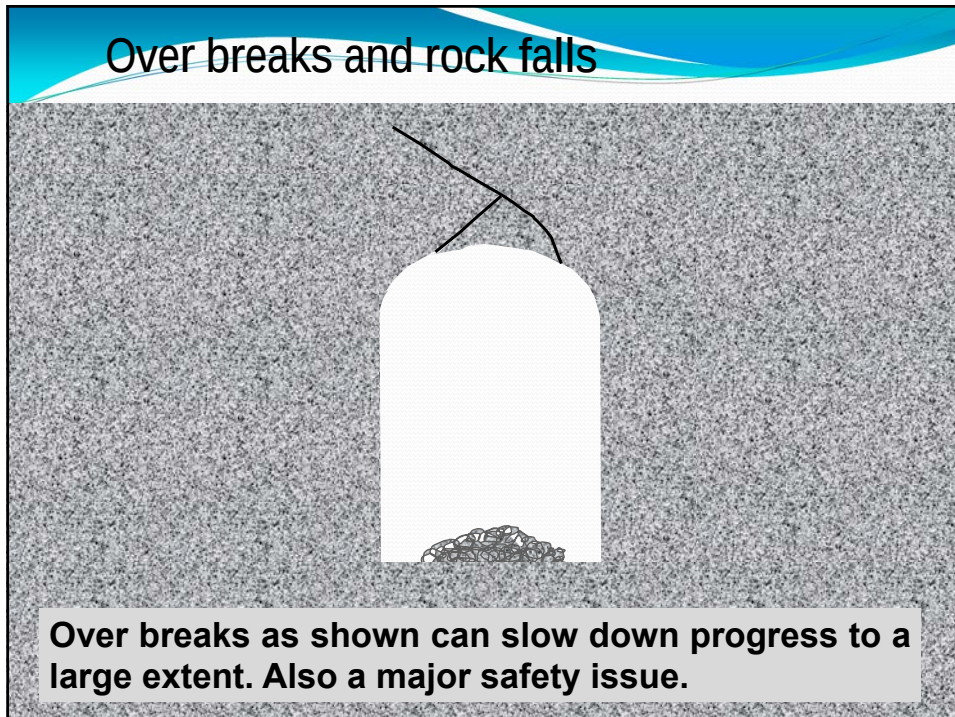
- face blasting with horizontal drillholes for tunnelling or top heading excavation,
- benching with horizontal drillholes, or
- benching with vertical drillholes.

Design and Construction of Caverns

A good rock tunnelling practice can be achieved by:

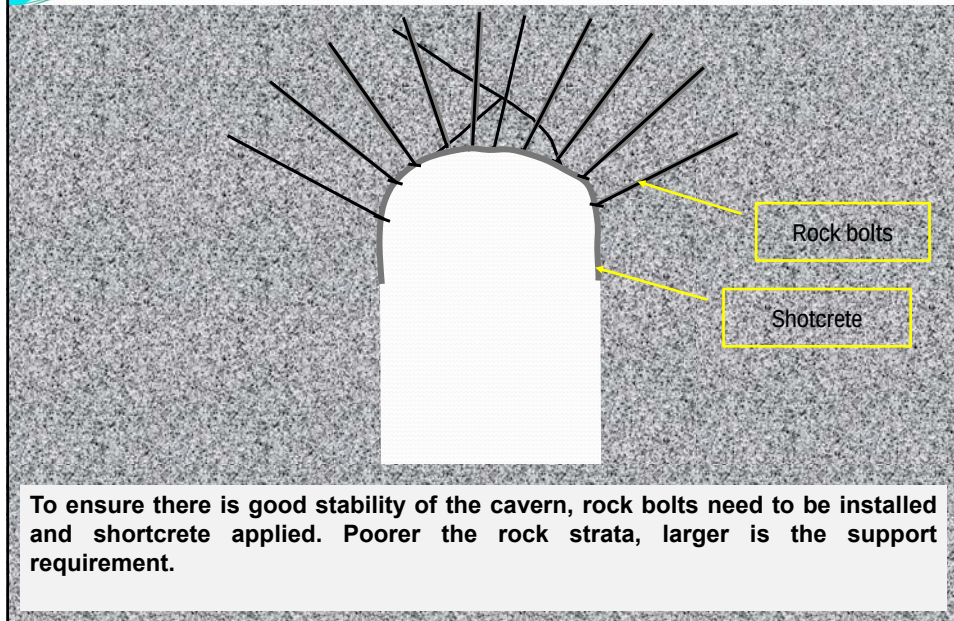
- Good knowledge of rock properties through appropriate site investigation;
- Good rock mechanics analysis, including using physical and numerical modelling, to anticipate the response of rock mass during and after construction;
- Good engineering practice supported by monitoring and risk control exercises.

Over breaks and rock falls

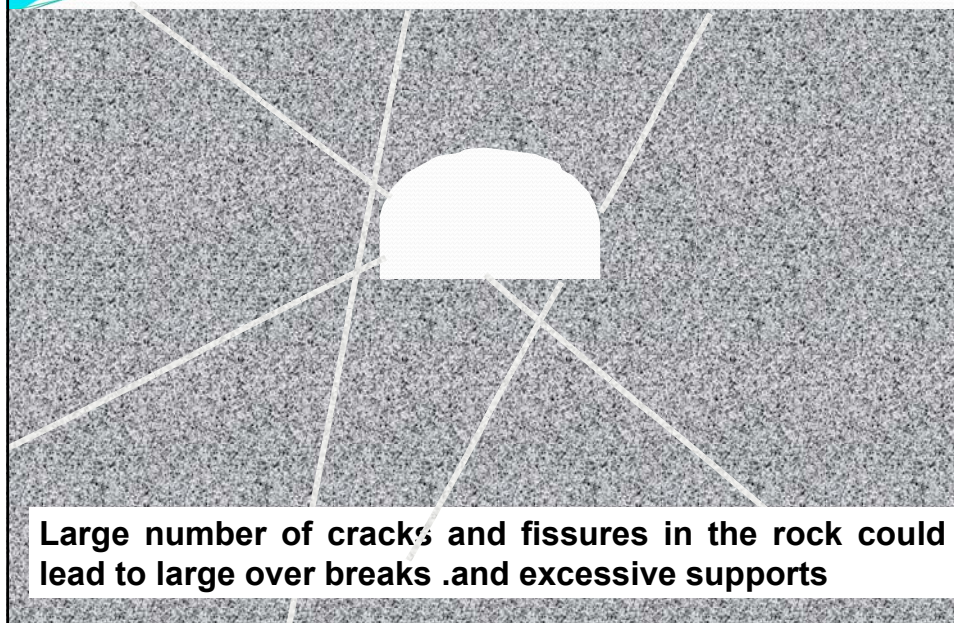


Over breaks as shown can slow down progress to a large extent. Also a major safety issue.

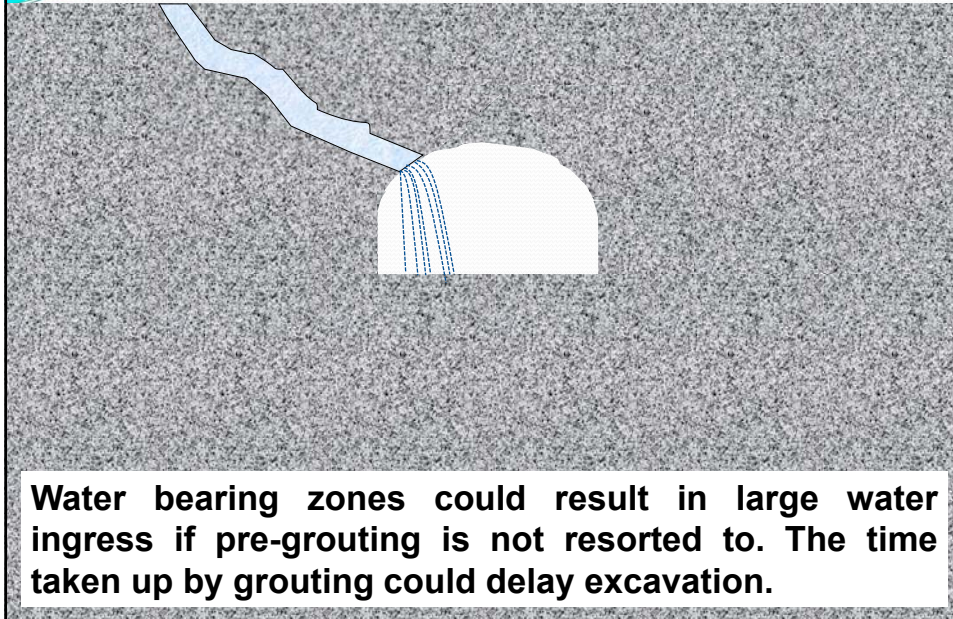
Method of addressing poor rock conditions



Example of Poor Rock Conditions

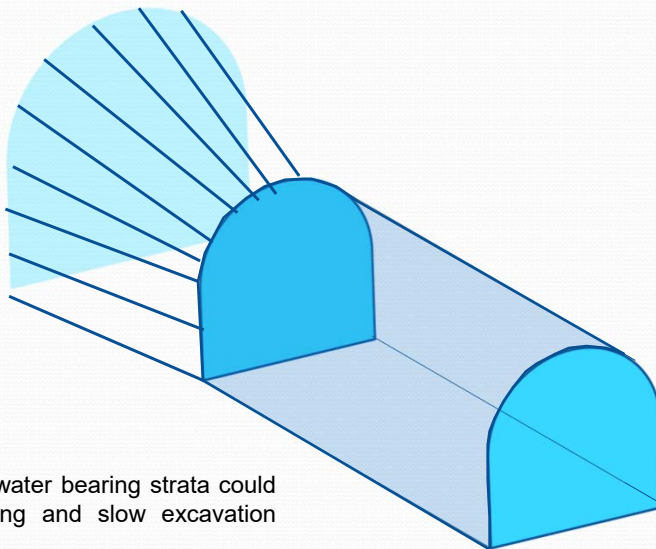


Example of a water bearing zone

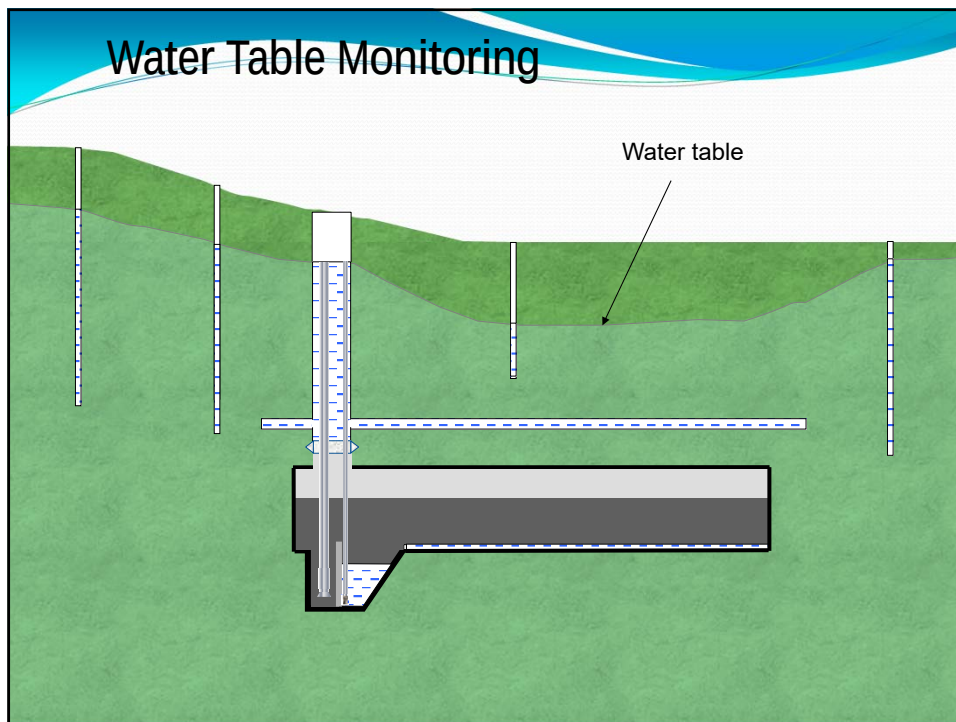


Pre-Grouting in excessive water bearing zones

Pre-Grouting involves drilling a number of holes on the face and pumping cement paste under pressure into them till sealing is achieved.



High permeability and water bearing strata could result in higher grouting and slow excavation progress.



LNG – Liquefied Natural Gas

- LNG is the liquid form of the natural gas people use in their homes for cooking and heating.
- Natural gas is also used as fuel for generating electricity.
- Natural gas and its components are used as raw material to manufacture a wide variety of products, from fibers for clothing, to plastics for healthcare, computing, and furnishings.

LNG – Liquefied Natural Gas

- The most common use of LNG is for “peakshaving.” Peakshaving is a way local electric power and gas companies or utilities store gas for peak demand that cannot be met via their typical pipeline sources.
- The utility companies liquefy natural gas when it is abundant and available at off-peak prices, or they purchase LNG from import terminals supplied from overseas liquefaction facilities.

LNG – Liquefied Natural Gas

- When gas demand increases, the stored LNG is converted from its liquefied state back to its gaseous state, to supplement the utilities pipeline supplies.
- LNG is also currently being used as an alternative transportation fuel in public transit and in vehicle fleets such as those operated by many local natural gas utilities companies for maintenance and emergencies.

LNG – Liquefied Natural Gas

- Natural gas comes from reservoirs beneath the earth's surface. Sometimes it occurs naturally and is produced by itself (non- associated gas), sometimes it comes to the surface with crude oil (associated gas), and sometimes it is being produced constantly such as in landfill gas.
- Natural gas is a fossil fuel, meaning that the natural gas we produce from the subsurface is derived from organic material deposited and buried in the earth millions of years ago. Other fossil fuels are coal and crude oil.

LNG – Liquefied Natural Gas

- Liquefied natural gas (LNG) is natural gas that has been cooled to the point that it condenses to a liquid, which occurs at a temperature of approximately - 256°F (-161°C) at atmospheric pressure.
- Liquefaction reduces the volume of gas by approximately 600 times thus making it more economical to store natural gas where other forms of storage do not exist, and to transport gas over long distances for which pipelines are too expensive or for which other constraints exist.
- Liquefaction makes it possible to move natural gas between continents in specially designed ships.

LNG – Liquefied Natural Gas

The major stages of the **LNG value chain**, excluding pipeline operations between the stages, consist of the following.

- **Exploration** to find natural gas in the earth's crust and **production** of the gas for delivery to gas users.
- **Liquefaction** to convert natural gas into a liquid state so that it can be transported in ships.
- **Shipping** the LNG in special purpose vessels.
- **Storage** of LNG in specially made tanks, and **regasification** to convert the LNG from the liquid phase to the gaseous phase, ready to be moved to the final destination through the natural gas pipeline system.

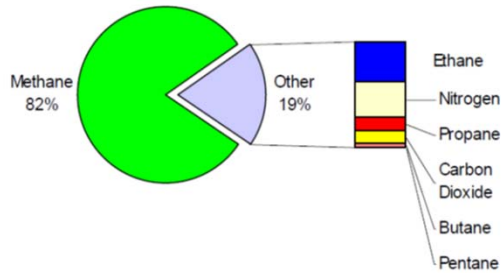
LNG Value Chain



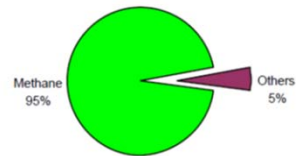
The LNG “value” or “supply” chain consists of four highly linked, interdependent segments - exploration and production (or E&P); liquefaction; shipping - from the point of liquefaction to the final destination; and receiving, storage and regasification at the final destination. We use the term “value” because at each stage investments are made to take natural gas from an unusable state to one in which optimal use of natural gas as a critical energy fuel and feedstock for materials can be achieved.

Typical Natural Gas and LNG Composition

Typical Natural Gas Composition



Typical LNG Composition



LNG – Liquefied Natural Gas

During liquefaction, contaminants found in produced natural gas are removed to avoid freezing up and damaging equipment when the gas is cooled to LNG temperature (-256°F) and to meet pipeline specifications at the delivery point. The liquefaction process can be designed to purify the LNG to almost 100 percent methane.

LNG – Liquefied Natural Gas

The liquefaction process entails cooling the clean feed gas by using refrigerants. The liquefaction plant may consist of several parallel units (“trains”). By liquefying the gas, its volume is reduced by a factor of 600, which means that LNG at -256°F uses 1/600th of the space required for a comparable amount of gas at room temperature and atmospheric pressure.

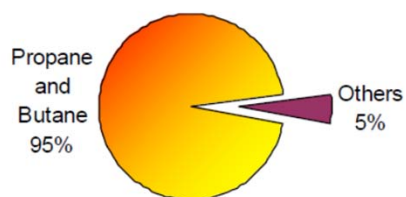
LNG – Liquefied Natural Gas

- LNG is a cryogenic liquid. The term “cryogenic” means low temperature, generally below -100°F. LNG is clear liquid, with a density of about 45 percent the density of water.
- At both liquefaction and receiving and regasification facilities, the LNG is stored in double-walled tanks at atmospheric pressure. The storage tank is really a tank within a tank. The annular space between the two tank walls is filled with insulation. The inner tank, in contact with the LNG, is made of materials suitable for cryogenic service and structural loading of LNG. These materials include 9 percent nickel steel, aluminum and pre-stressed concrete. The outer tank is generally made of carbon steel or pre-stressed concrete.

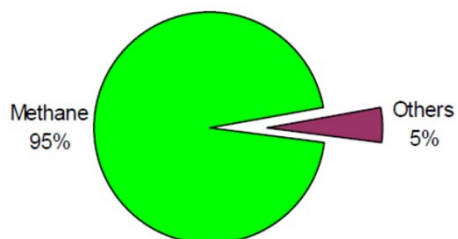
LPG – Liquefied Petroleum Gas

- LNG is not the same as **Liquefied Petroleum Gas (LPG)**. LPG is often incorrectly called propane. In fact, LPG is predominantly a mixture of propane and butane in a liquid state at room temperatures when under moderate pressures of less than 200 psig (pounds per square inch gauge) (psig) (1378.96 kPa).
- Worldwide, LPG is used heavily for domestic purposes such as cooking and heating water.

LPG Composition



LNG Composition



CNG – Compressed Natural Gas

LNG is not the same as **compressed natural gas (CNG)**. CNG is natural gas that is pressurized and stored in welding bottle-like tanks at pressures up to 3,600 psig. Typically, CNG is the same composition as pipeline quality natural gas, i.e., the gas has been dehydrated (water removed) and all other elements reduced to traces so that corrosion is prevented. CNG is often used as a vehicle transportation fuel and is delivered to an engine as low-pressure vapor (up to 300 psig). CNG is often misrepresented as the only form of natural gas that can be used as vehicle fuel. However, LPG and LNG are also common transport fuels.

LNG Storage in Underground Lined Rock Caverns

- Since natural gas is imported in the liquid phase by ships, the refrigerated storage of LNG is more economical than the conventional storage in the form of gas.
- At LNG terminals, conventional aboveground tank or in-ground tank type has been successfully used worldwide to store LNG. Nevertheless, these types of the storage tanks are less attractive as they require large land area for storage. Large reclamation of seaside and industrial area has already progressed and free remaining area is very small and expensive.

LNG Storage in Underground Lined Rock Caverns

- Based on experience of underground storage for crude oil and various types of hydrocarbons, underground storage system was thought to be more economical way to store LNG regardless of the above-mentioned conditions.
- Many attempts have therefore been made in the past to store LNG underground in unlined containment, though without success (Anderson, 1989; Dahlström and Evans, 2002).

LNG Storage in Underground Lined Rock Caverns

- One of the most significant problems related to the underground storage of cryogenic material is the need to prevent leakage of liquid and gas from the containment system to the rock mass caused by tensile failures due to shrinkage of the rock mass around the caverns (Monsen and Barton, 2001).
- The failures of underground storage caverns were due to thermal stresses generating cracks in the host rock mass.

LNG Storage in Underground Lined Rock Caverns

- The thermal cracks induced by extremely low temperatures in the rock mass contributed to the deterioration of the operational efficiency. Gas leakage and increased heat flux between the ground and storage occurred (Dahlström, 1992; Glamheden and Lindblom, 2002).

LNG Storage in Underground Lined Rock Caverns

- The way to prevent a hard rock mass from cracking at LNG boiling temperatures (-162°C) is to locate the unlined storage cavern deep enough below ground level so that the geostatic stresses counterbalance the tensile stresses caused by the cooling. The necessary depth varies with rock type from 500 to 1000m, which renders this unlined cavern storage concept very expensive (Amantini and Chanfreau, 2004; Amantini et al., 2005).

LNG Storage in Underground Lined Rock Caverns

- New concept of storing LNG in a lined rock cavern (LRC) with containment system which can overcome these problems has been developed by Geostock, SKEC and SN Technigaz with the help of KIGAM (Korea Institute of Geoscience and Mineral Resources).

Concept of a lined rock cavern for LNG storage

- New concept of storing LNG in a lined rock cavern (LRC) with containment system which can overcome these problems has been developed by Geostock, SKEC and SN Technigaz with the help of KIGAM (Korea Institute of Geoscience and Mineral Resources).
- The basic concept for the development of LNG storage in membrane lined rock cavern is a combination of well-proven technologies and a new concept named as “formation of ice ring”

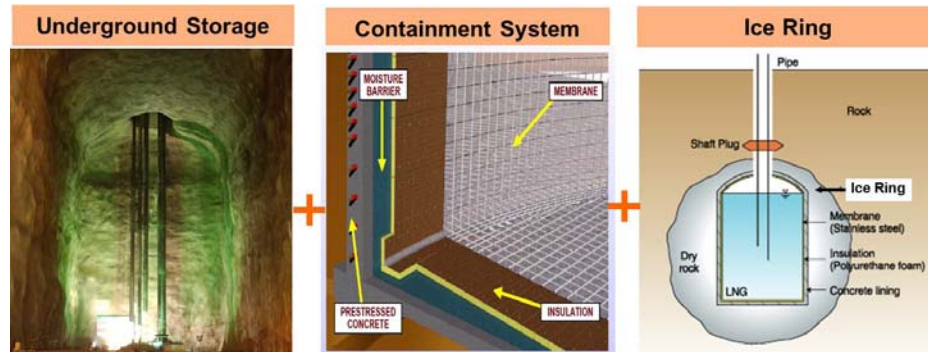
Concept of a lined rock cavern for LNG storage

- The concept consists in protecting the host rock against the extreme low temperature and in providing a liquid and gas tight liner using insulating panels fixed on a concrete lining and a corrugated stainless steel membrane. Similar containment systems are used in LNG carriers since 30 years without any troubles.

Concept of a lined rock cavern for LNG storage

- The thermal characteristics and thickness of the insulation is designed in order to achieve allowable minimum temperature in the rockmass for the design life of the storage and boil-off rate less than 0.1% per day.
- A dedicated water drainage system made of boreholes drilled from the surface and/or dedicated drainage galleries installed around the cavern allows controlling the hydrostatic pressure and the ice formation in the rock mass during the cooling down process.
- Process and equipment to operate the storage are similar to aboveground tanks.

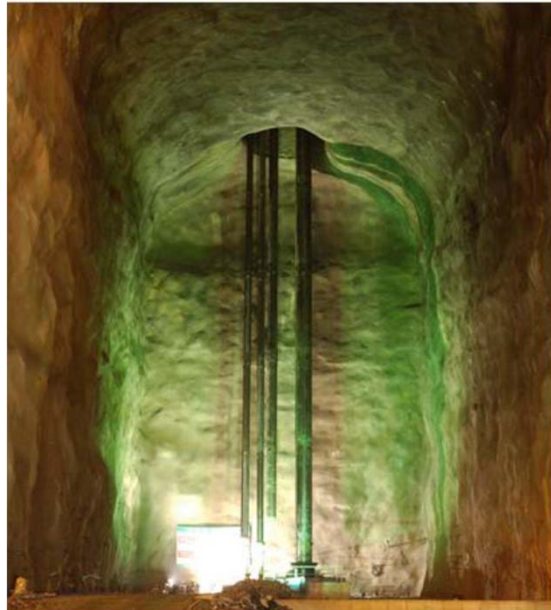
Concept of a lined rock cavern for LNG storage



Underground storage cavern

- The underground storage technology, which proved reliable, experienced a strong development related to its intrinsic qualities but also on the innovations and technical progress from which it regularly profited.
- Currently, large capacity of storage cavern units with more than several hundreds of thousands of cubic meter, for a broad range type of hydrocarbon products and crude oil, are constructed and operated successfully.

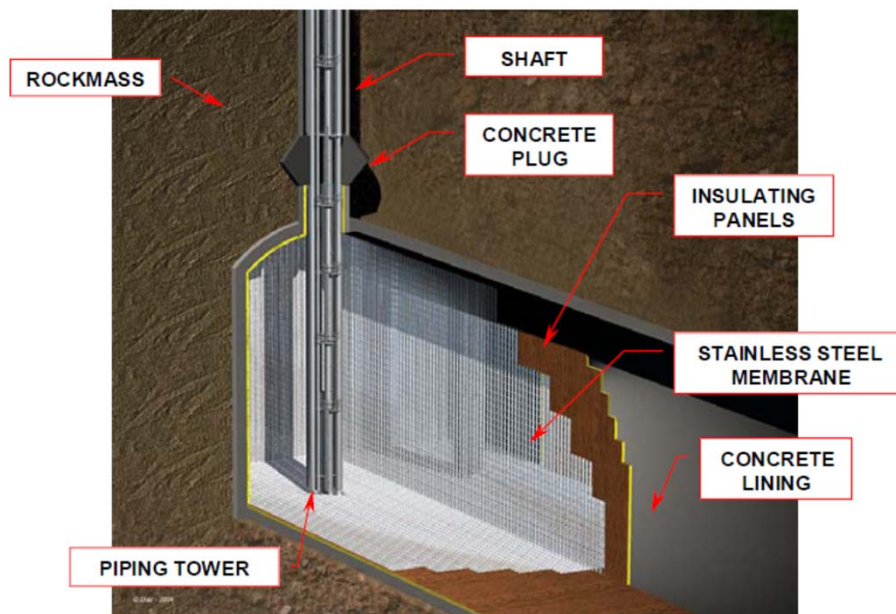
Underground Storage



Membrane containment system

- The membrane containment system, which was introduced in 1962 on a prototype ship, has since been successfully used in several LNG carriers and storage tanks.
- This membrane system provides the proper thermal protection to the surrounding rock mass, preventing excessive stresses and crack formation and reducing boil-off to level comparable with conventional LNG storage tanks.

Membrane containment system



Ice ring and drainage system

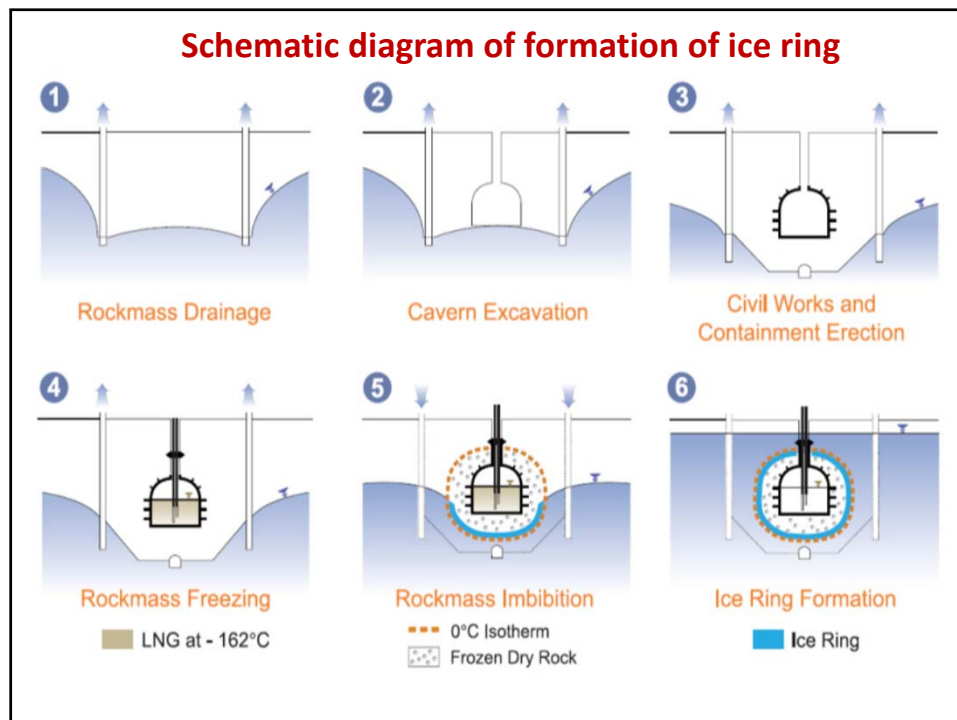
- Formation of ice ring as an impervious second barrier against the leakage of contained LNG.
- Groundwater is temporarily removed from the rock surrounding the cavern during the first phase of construction. This preliminary de-saturation of the host rock mass aims at preventing unacceptable hydrostatic pore pressure and ice formation behind the cavern lining.

Ice ring and drainage system

- After excavation of cavern and installation of containment system are completed, LNG will be stored in the cavern and then the cold front starts propagation from the cavern at the same time.
- When the cold front has advanced far enough from the cavern wall, drainage can be stopped to allow groundwater progressively to rise up and quickly form a thick ring of ice around the cavern.

Ice ring and drainage system

- After ice ring is completely formed, operation of the drainage system is stopped.
- The drainage period during LNG storage will last several months or years depending on the thermal properties of rock masses and hydro-geological characteristics on the site.

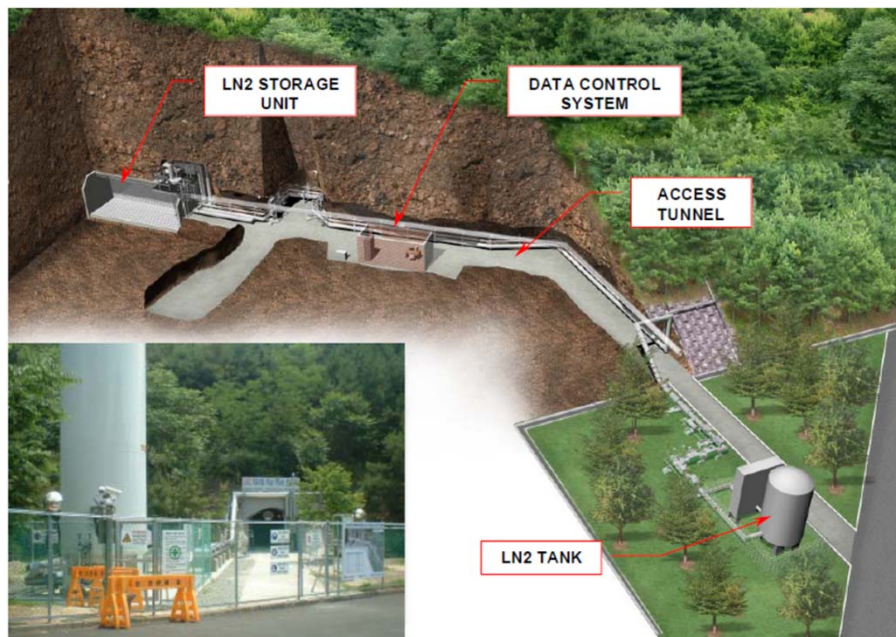


LNG pilot plant

The objectives of this pilot project are as follows:

- To demonstrate the feasibility of the lined rock cavern concept,
- To check adequacy between the results of previous computations and laboratory tests regarding in-situ measurements,
- To test the overall performance of the storage and
- To gain sufficient experience to improve design and construction of industrial scale projects.

LNG pilot plant



LNG pilot plant

- The host hard-rock is a fissured granite of fair to good quality. Access to the pilot cavern is provided through an existing horizontal tunnel and experimental cavity roof lies at a depth of about 20 m from the ground. The overall cavern working volume is 110 m³.

LNG pilot plant

The cavern was filled with liquid nitrogen (LN2) and operated for one year (6 months cooling and 6 months thawing).

LN2 is safer than LNG but also more challenging with a boil off temperature of -196°C at atmospheric pressure. The insulation layer has been limited to 10 cm of PU foam to speed the cooling down and the spread of the cold front in the rockmass.

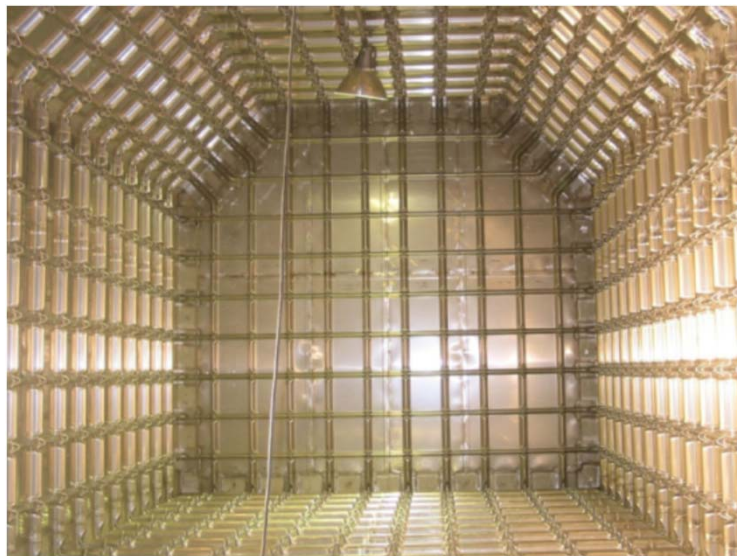
Membrane containment system

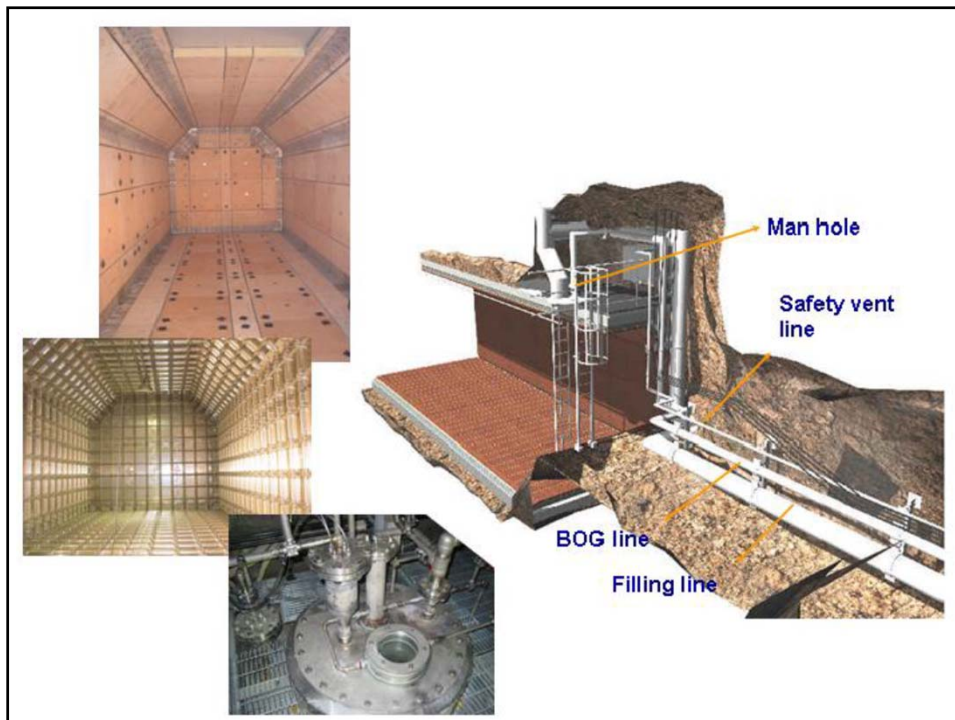
- The containment system is composed of several layers, from rock to LNG.
- For the Daejeon LNG pilot cavern, insulating panels made of foam are sandwiched between plywood sheets and are bonded on the concrete using load-bearing mastic. The insulation panel thickness is 300 mm to ensure that the rock temperature will not fall below -50°C after 30 years and the boil-off rate will stay at an acceptable limit.

Membrane containment system

- Finally, a 1.2 mm thick stainless steel corrugated membrane attached to the insulation panel provides gas tightness at a low temperature. All the surfaces (e.g. bottom, walls, chambers and vault) are covered with concrete lining, insulating panels and the stainless steel membrane sheets.

Containment system used for the pilot test





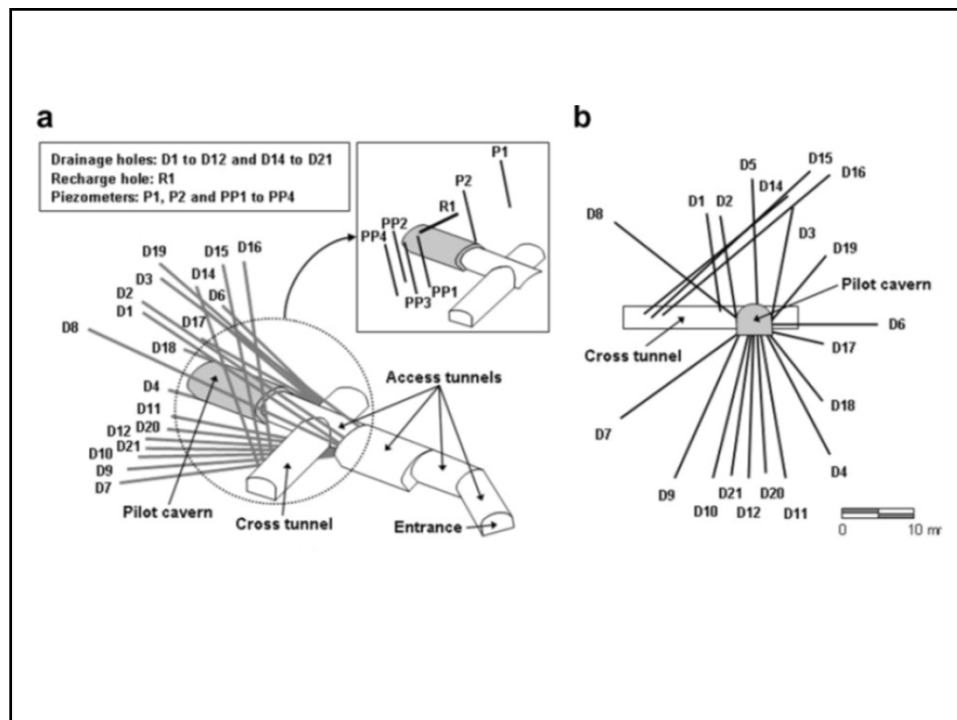
Drainage system

The purpose of the drainage system is

- to reduce water entry in the cavern during concrete casting,
- to reduce humidity percolating through the concrete during the installation of the containment system,
- to drain the rock mass and maintain a low water saturation degree in the surrounding rock mass during the cooling phase, and
- to control the re-saturation of the host rock mass once the frozen area around the cavern reaches a suitable thickness.

Drainage system

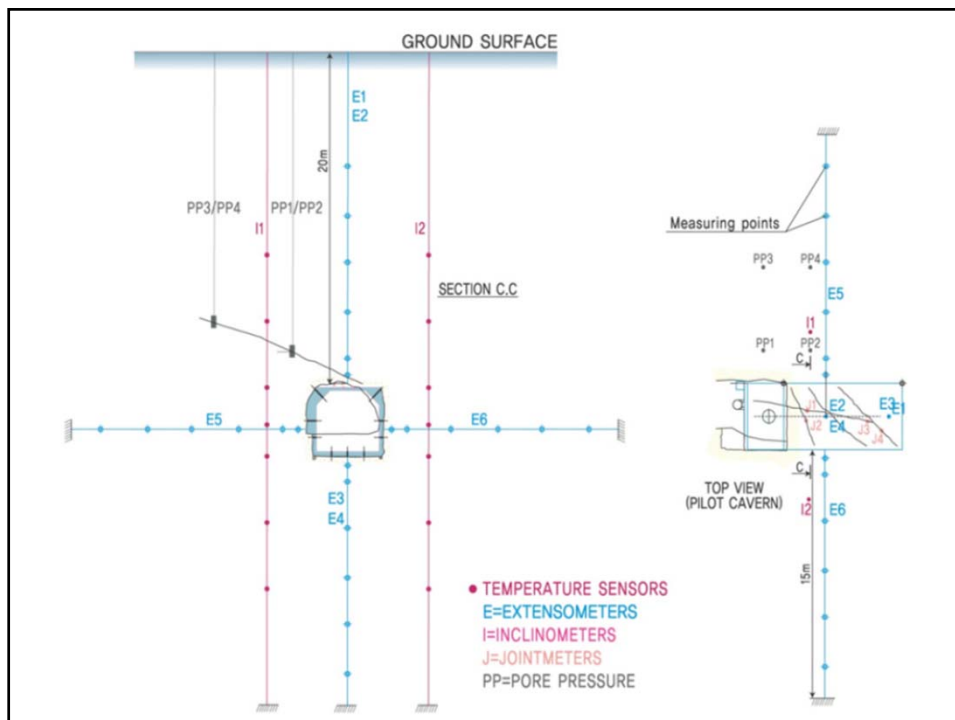
- Before the cavern was excavated, 21 boreholes were drilled from the tunnel drifts for drainage near the cavern walls, and the water table was lowered to 8 m below the cavern floor.



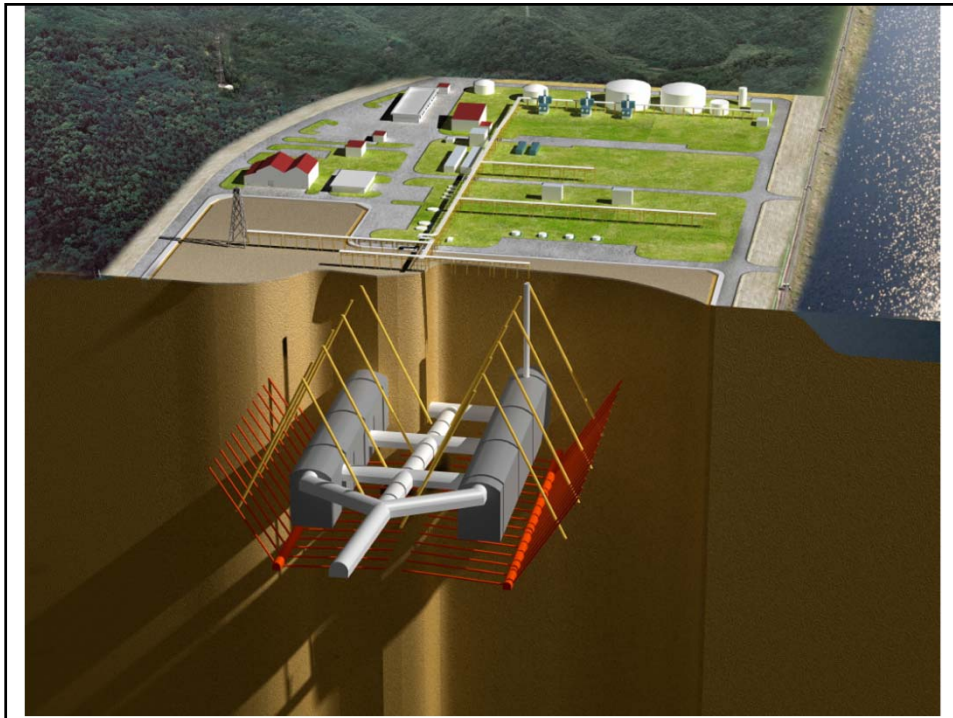
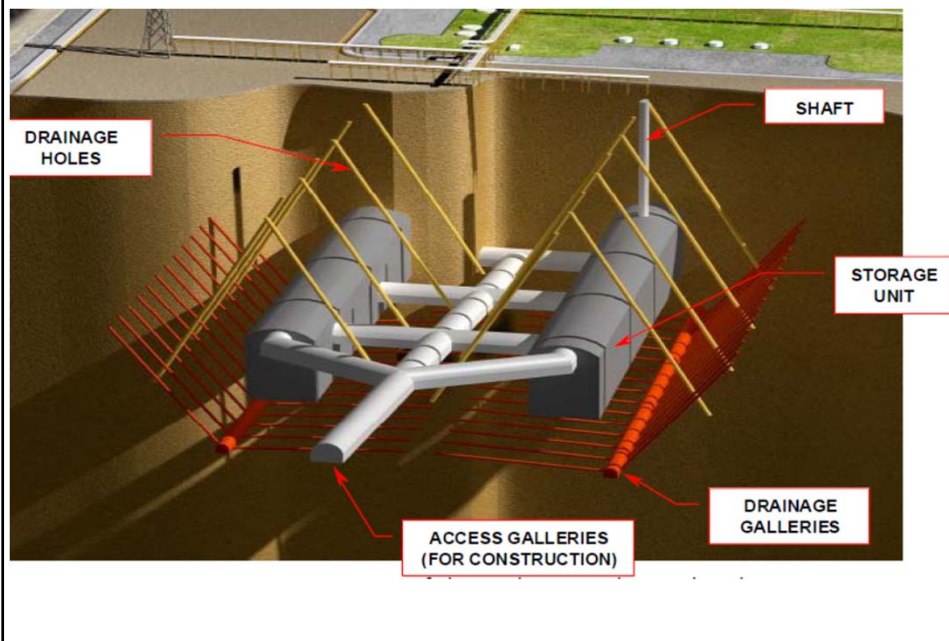
Monitoring system

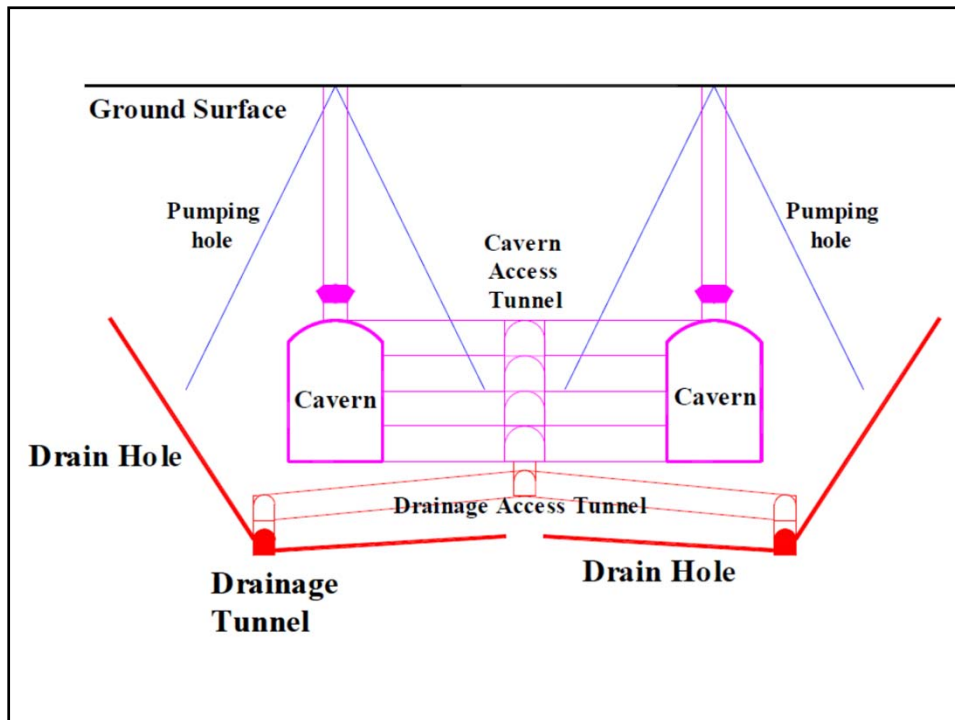
Instrumentation was installed in containment system and cavern to follow main parameters (inner pressure and temperature, LN2 level, containment deformation) and to operate the pilot.

- The pilot cavern and its surrounding rockmass are equipped with a full set of geotechnical instruments which allows monitoring of:
 - temperature profiles around cavern,
 - induced displacements in the concrete structure (strain gages) and the rock mass (extensometers and inclinometers),
 - induced joint opening (crackmeters),
 - pore pressure in fissures,
 - drainage and seepage flowrates,
 - water table level



Typical underground storage system





Shape of storage cavern and tunnel

The general cross section of storage gallery is comparable to unlined underground storage technology, with horse shape galleries. Dimension of the gallery is typical 30 m height by 20 m width, which can be adapted to rock conditions. The length of cavern can be from 150 m up to 270 m, which depends upon the unit capacity per gallery.

Advantages and economics of underground LNG storage

There are many advantages of underground storage in terms of safety, security and environmental acceptability compared with aboveground tanks and in-ground tanks:

Safety: Underground storage is much safer in the case of fire on plant premises or decreased potential damages in the case of nearby industrial accidents, on account of its multi-component barrier with liner and ice ring, and lower vulnerability against earthquake and typhoon.

Advantages and economics of underground LNG storage

Security: Enhances security, as underground storage can easily prevent sabotage or terrorism acts.

Environmental impact: As there is no need of large reclaimed areas and less earthwork at ground level, underground storage is environmentally more friendly, and is readily accepted by people located nearby.

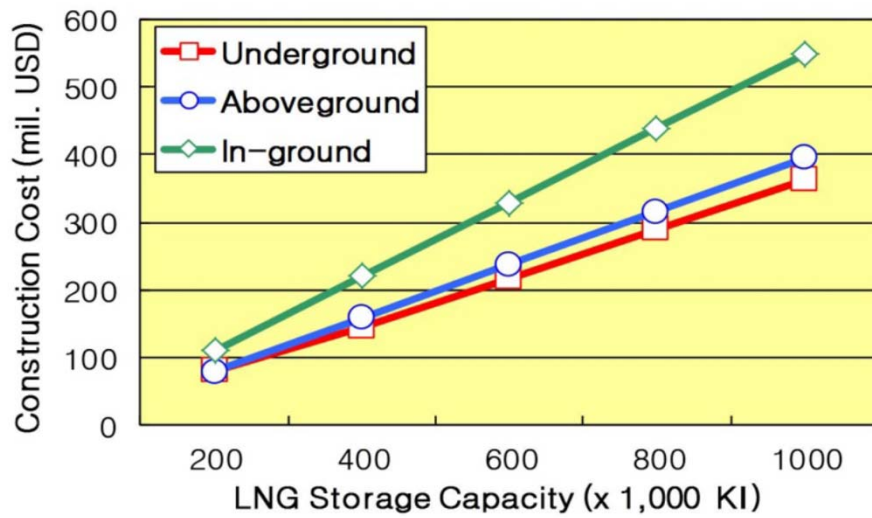
Advantages and economics of underground LNG storage

Use of land space: Land of aboveground can be used better because of the minimization of total space required for the LNG terminal. This is due to the fact that LNG storage is about 50 m underground, this can represent a huge cost saving especially in seashore areas where industries are already developed and free remaining areas are small and expensive.

It is also the case in areas whose topography needs expensive reclaimed land.



Required area at grade depending on storage type for 320,000 m³ (Orange colour means a storage space, purple colour means the required aboveground space)



STORAGE PRINCIPLES OF ROCK CAVERNS FOR LPG

To reduce the volume of the LPG-storage, the gas has to be stored as a liquid.

There are two methods used to make the gas into a liquid, by reducing the temperature or increasing the pressure.

The most practical way for rock storage is to reduce the temperature to approximately -30°C at which point the LPG is liquid.

The LPG storage needs a special refrigeration unit for maintaining the temperature.

STORAGE PRINCIPLES OF ROCK CAVERNS FOR LPG

The storage of LPG is basically made on the same main principles as for oil products.

The ground water is used as the seal for tightening the rock masses.

The difference will be that due to the cold temperatures the water will freeze to ice.

STORAGE PRINCIPLES OF ROCK CAVERNS FOR LPG

A special cool-down activity is also required for preparing the storage for the first import of LPG.

To seal off all water leakages in LPG storages during the cool-down period is fundamental for operation and lifetime.

The leakages must be small as the heat energy in the water in a big leakage might prevent it from freezing during the cool-down period.

STORAGE PRINCIPLES OF ROCK CAVERNS FOR LPG

If the leakages are not sealed off during the cool-down period, water continues to seep into the cavern the water will freeze inside the cavern and cannot be pumped out.

Due to this build up of ice the LPG pumps can be blocked off and the available storage volume reduced. After some time, the storage can be considered as lost.

STORAGE PRINCIPLES OF ROCK CAVERNS FOR LPG

INERTING OF CAVERNS PRIOR TO START-UP

For the LPG storage the cavern and surrounding rock mass was cooled down with air. Inspections were performed twice a week to monitor the movements in the rock mass via mini extensometers.

Observations of any ice formations were performed and noted for controlling when water leakages, which caused ice formations, were frozen. The time and energy needed was in advance carefully calculated to optimise the cooling equipments, both for the cool-down period and for the permanent operation of the storage.

DESIGN PROPERTIES (LPG)

In principle the construction of unlined underground caverns for LPG can be accomplished where the rock formation is acceptable for excavation of caverns. The quality of the bedrock and the groundwater conditions are obviously one of the major factors affecting the construction cost.

Thus, rock caverns for oil products and LPG.s are feasible to construct wherever the bedrock is reasonably tight, stable and un-faulted and where there is a relatively stable groundwater level near the ground level.

DESIGN PROPERTIES (LPG)

During the design of underground openings, investigations on geological and hydrogeological conditions are of most importance. The information gained is a requisite in order to design and optimize, at the selected location, opening orientation and shape, excavation method and adequate support to secure long term stability etc, referring to safety, environmental and economical considerations.

In the design process of underground hydrocarbon storages, additional design criteria must be considered. The main additional criteria is for minimum groundwater levels, groundwater pressures and the required storage condition for the product.

DESIGN PROPERTIES (LPG)

For refrigerated storages, special considerations regarding rock stress redistribution due to temperature changes and consequently the adequate support system required for stable conditions. Depending on which hydrocarbon product to be stored, different design criteria must be fulfilled in order to secure containment and protect the environment and product from contamination.

Crude oil and some refined products may be stored at atmospheric pressure in unlined caverns; the containment is secured by the groundwater that prevents the product from penetrating the rock mass.

DESIGN PROPERTIES (LPG)

Petroleum gases are stored in liquid state as (LPG), either in a compressed or a refrigerated storage.

In the **compressed storage** the containment is secured by the groundwater pressure, and

In the **refrigerated storage** by the ice that forms around the cavern.

STORAGE CONDITION OF LPG

Liquid petroleum gases (LPG) are mixtures of butane and propane with varying composition and properties.

Pure propane may be stored at -42°C (boiling point) at atmospheric pressure in a fully refrigerated storage. Any temperature increase will induce boiling, which will continue until the pressure reaches a value that matches the temperature on the vapor pressure curve for propane. For example, if the temperature were to rise to -20° , boiling would commence until a storage pressure of 0.2 MPa above atmospheric is reached. If no refrigeration is added to the storage, the temperature will rise to the ambient rock temperature.

STORAGE CONDITION OF LPG

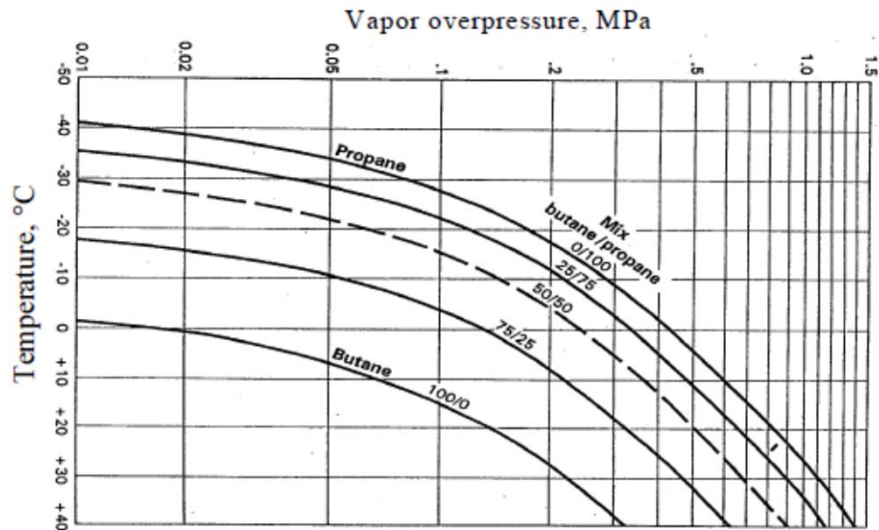
Storages at atmospheric pressure are commonly called “fully refrigerated”. and storage without refrigeration, i.e. at ambient temperature, is commonly named “fully compressed storage”.

Fully refrigerated butane storages operates at -2°C for atmospheric pressure and fully compressed storage operates at 0.05 to 0.2 MPa overpressure, depending on the temperature.

However, unless its not absolutely clear that only butane will be stored, any LPG storage should be designed for propane conditions.

From Figure, it is evident that LPG may be liquefied readily at slight pressurization.

Relation between temperature and vapor overpressure for LPG



DESIGN PRINCIPLES FOR UNLINED LPG CAVERNS

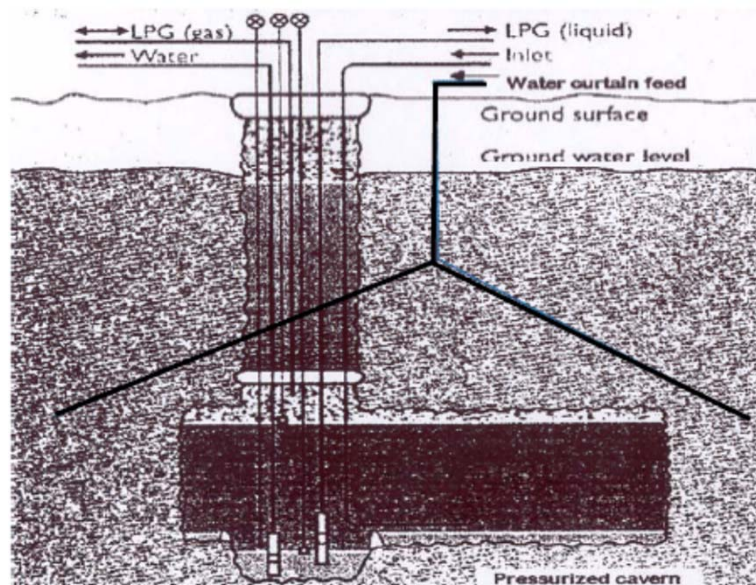
Compressed caverns:

The principle of compressed gas storages in unlined rock caverns is based on the containing effect of the ground water in the host rock.

In order to assure containment, the pressure of the gas cushion formed on top of the LPG must at all time be less than the actual groundwater pressure close to the cavern walls and roof and the water pressure must increase away from the cavern.

This entails that a certain inflow of groundwater into the cavern must be allowed. One way of assuring such conditions is to install an array of drill holes (a water curtain) outside the cavern periphery. The holes are then continuously pressurized with water at a controlled pressure.

Conceptual design of a compressed LPG storage cavern



DESIGN PRINCIPLES FOR UNLINED LPG CAVERNS

Compressed caverns:

The necessity of water curtains and the key issues:

- Which type of rock require water curtain?
- Should the water curtain be pressurized before excavating the cavern beneath?
- What hydraulic gradient is required to prevent gas entry and/or migration from the cavern?
- What quantity of water will need to be injected
- How will the water curtain affect the flow into the cavern and, consequently the dissolution of gas?
- What will be the long-term effects of a water curtain, e.g., on hydro-chemical reactions?

DESIGN PRINCIPLES FOR UNLINED LPG CAVERNS

Refrigerated caverns:

In a fully refrigerated storage very small vapor pressures will be acceptable and the product is stored at a temperature close to its atmospheric boiling point. The rock mass that surrounds the storage has a considerable higher temperature, and physical contact between the LPG and the rock shall only occur at the cavern surfaces. If LPG were allowed to enter into to the rock mass fracture system, it will reach a higher temperature and the gas will vaporize and consequently increase the .boil-off.

DESIGN PRINCIPLES FOR UNLINED LPG CAVERNS

Refrigerated caverns:

However, the frozen groundwater in the rock mass will prevent LPG from migrating into the rock mass. Therefore it is important to maintain groundwater saturation in the rock mass, during construction and initial cooling.

During the initial cooling, groundwater will continue to enter the cavern system and it must be possible to pump it out. When the frozen zone has started to develop, the water seepage will cease. As cooling proceeds, the thickness of the frozen zone will grow until a steady state is reached, at which the cooling power added to the cavern exactly matches heat influx from the rock mass.

DESIGN PRINCIPLES FOR UNLINED LPG CAVERNS

Refrigerated caverns:

At the top of the cavern there will be a slight pressure. The tendency of the gas to migrate upwards is restricted by the ice-filled fractures. However, even though gas leakage are restricted by the ice-filled fracture system around the cavern, there is no guarantee that the ice-walls are perfectly gas-tight at all times.

The stress redistribution, due to thermo-mechanical loading and continues unloading of the rock mass, may create new passages for gas. A design criterion is therefore that free water must exist above the frozen zone covering the cavern roof, i.e. the 0°-isotherms at steady state must be located below the groundwater level.

DESIGN PRINCIPLES FOR UNLINED LPG CAVERNS

Refrigerated caverns:

In successful underground gas storages, the boil-off must be controlled, which requires a perfectly sealed rock cavern. Consequently the heat inflow is in the form of conduction. If liquefied gas is allowed to leak out in rock discontinuities, it will warm up and vaporize.

The vaporized gas that returns into the rock cavern transport heat from the surrounding rock and the heat flux will increase dramatically due to convection.

DESIGN PRINCIPLES FOR UNLINED LPG CAVERNS

Refrigerated caverns:

In principle, as long as compressive stresses act over a fracture, the fracture will remain closed and sealed by ice, and the heat flux is controlled by conduction only.

Due to thermal contraction of the rock mass the tangential rock stresses around the cavern will decrease. When full stress relaxation is obtained, at a certain critical temperature, free contraction will develop, i.e. the fractures may start to open and allow for possible gas migration into the rock mass (Dahlström, 1992).

DESIGN PRINCIPLES FOR UNLINED LPG CAVERNS

Refrigerated caverns:

An obvious design criterion could be to locate the cavern at such location that thermally induced stresses will be lower than initial compressive rock stresses around the cavern. However, other factors as expansion of ice in fracture and ice pressure etc. is involved and therefore such a criterion would be quite over-conservative in terms of required depth for LPG storages.

Experience from existing refrigerated storages has shown that the groundwater criterion will be sufficient to ensure tight caverns.

DESIGN PRINCIPLES FOR UNLINED LPG CAVERNS

Refrigerated caverns:

The thermally induced stress relaxation will decrease or even vanish the initial arching effect in the roof and rock blocks may become unstable if not properly affixed by adequate support.

It is clear that the Thermo-mechanical behavior of the cavern is an important issue, and is in fact one of the key design factors.

Advantages of going underground

- Product is located at a depth and is fully isolated.
- External Fires will not affect Storage
- Maximum security. Safety hazards on account of sabotage, storms, earthquakes and explosions are minimized.
- Surface land requirement is low
- Caverns by their very nature require very low maintenance.
- Beyond certain capacities the capital cost per ton of storage is low.
- Environment benefits. The rock debris generated can be used for infrastructure development.